A Sample Article Using IEEEtran.cls for IEEE Journals and Transactions

IEEE Publication Technology, *Staff, IEEE,*

***Abstract*—This paper studies online multivariate regional load forecasting under delayed feedback and distribution shift. We develop a plug-and-play forecasting framework that combines invariant representation learning, factor-based expert modeling, and a three-stage online controller. The framework separates stable predictive structure from variant fluctuations, performs relevance-guided sparse updates only when experts become unstable, and applies transient drift compensation at inference time. Extensive experiments on multiple load forecasting datasets and multiple transformer-based backbones show that the proposed framework consistently improves over frozen deployment and representative online baselines under shifted online streams.**

***Index Terms*—online load forecasting, multivariate time series, distribution shift, invariant learning, transformer backbone, delayed feedback.**

I. Introduction

Regional load forecasting in real deployment environments is inherently nonstationary. Seasonal transitions, weather fluctuations, and changing demand patterns cause the online test stream to deviate from the distribution seen during offline training. In such settings, a strong offline forecaster alone is often insufficient, while naive online fine-tuning can be unstable under delayed labels and concept drift. This motivates an online forecasting framework that can preserve stable mechanisms, localize unstable components, and adapt conservatively when the distribution shifts.

The method developed in this paper follows that principle. We first isolate invariant predictive structure through environment-aware representation learning, then model dynamic load-generation mechanisms through a factorized multi-expert forecasting backbone, and finally perform structured online adaptation through Stage I, Stage II, and Stage III. Stage I recalibrates the factor mixture under delayed labels, Stage II performs relevance-guided sparse expert updates when instability is detected, and Stage III applies transient correction based on the discrepancy between variant and invariant representations.

Unlike methods tied to a single predictor, the proposed design is formulated as a plug-and-play framework. The same online controller can be combined with multiple transformer-based backbones, which allows the experiments to evaluate both forecasting strength and backbone generality. The following sections present the problem formulation, the method, and the empirical validation.

This paper was produced by the IEEE Publication Technology Group. They are in Piscataway, NJ.

Manuscript received April 19, 2021; revised August 16, 2021.

II. Preliminaries

A. Notation

Let $\{\mathbf{x}_t\}_{t=1}^{T}$ denote a multivariate time series, where $\mathbf{x}_t \in \mathbb{R}^N$ and N is the number of observed variables. We formulate time-series forecasting via a sliding-window scheme: for each reference time t, an input lookback window of length L is used to predict an output horizon window of length H. Specifically, we define the input window $\mathbf{X}_t \triangleq [\mathbf{x}_{t-L+1}, \ldots, \mathbf{x}_t] \in \mathbb{R}^{L\times N}$ and the output window $\mathbf{Y}_t \triangleq [y_{t+1}, \ldots, y_{t+H}] \in \mathbb{R}^{H}$, where y_t denotes the target variable. The formulation extends directly to multivariate targets by letting $\mathbf{Y}_t \in \mathbb{R}^{H\times d_{\text{out}}}$.

A forecasting model $f_\theta : \mathbb{R}^{L\times N} \to \mathbb{R}^{H}$ parameterized by θ maps past observations to future targets, and $\widehat{\mathbf{Y}}_t = f_\theta(\mathbf{X}_t)$. Given a pointwise loss function $\ell(\widehat{\mathbf{Y}}_t, \mathbf{Y}_t)$, standard training minimizes the empirical risk over the training set $\mathcal{D}_{\text{tr}} = \{(\mathbf{X}_t, \mathbf{Y}_t)\}_{t=1}^{N}$, formulated as: $\min_\theta \frac{1}{N}\sum_{t=1}^{N} \ell\big(f_\theta(\mathbf{X}_t), \mathbf{Y}_t\big)$.

B. Online Learning in OOD Settings

Let $P_{\text{tr}}(X, Y)$ and $P_{\text{te}}(X, Y)$ denote the joint distributions of input windows X and targets Y for training and test. OOD forecasting has $P_{\text{te}}(X, Y) \neq P_{\text{tr}}(X, Y)$ and can be characterized by the following shifts.

Covariate shift. $P_{\text{te}}(X) \neq P_{\text{tr}}(X)$; the input marginal changes, e.g., meteorological covariates differ across years or net-load profiles shift after large-scale renewable integration.

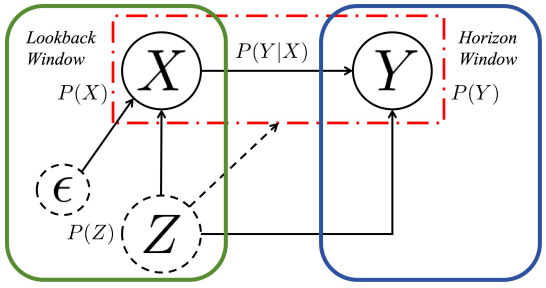
Label shift. $P_{\text{te}}(Y) \neq P_{\text{tr}}(Y)$; the target marginal changes, e.g., electrification increases mean load or industrial expansion alters peak frequency, causing bias under historical target density.

Conditional shift. $P_{\text{te}}(Y|X) \neq P_{\text{tr}}(Y|X)$; the mapping from history to future changes due to altered influence mechanisms among variables, e.g., demand response changes load sensitivity to temperature.

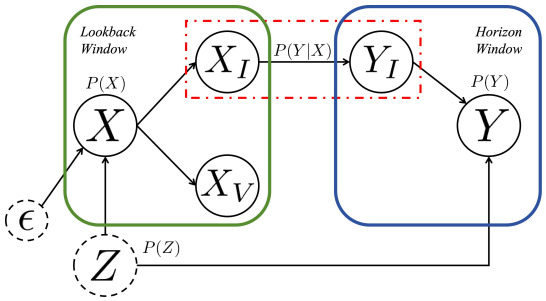
Latent shift induced by unobserved variables. Let Z denote unobserved drivers affecting both X and Y. Changes in $P_{\text{te}}(Z) \neq P_{\text{tr}}(Z)$ can induce joint shifts in observed variables. In load forecasting, such shifts may be triggered by extreme weather or public health emergencies, e.g., sudden demand changes beyond recorded covariates.

In practice, multiple shift types can co-occur, complicating OOD generalization. This motivates online learning in OOD settings to mitigate distribution drift by continually transforming newly observed series into training samples and updating model parameters to minimize forecast errors on recent data. However, conventional online learning faces three structural constraints: (i) due to the feedback delay in Section 1, updates can only use labels arriving at least L steps later; (ii) the online phase may encounter novel operating regimes outside

0000–0000/00$00.00 © 2021 IEEE



(a) **Conventional deep forecasting.** Conventional deep forecasters learn $P(Y|X)$ from historical data under $P_{\text{tr}}(X, Y)$. When an unobserved driver Z influences both X and Y, a deployment shift in $P(Z)$ can induce changes in the effective conditional $P(Y|X)$, thereby breaking the training-stage relationship and degrading performance.



(b) **Invariant learning module.**

Fig. 1. Examples of distribution shifts.

the historical distribution. (iii) some distribution drift is caused by variant unobserved factors. If updates are not applied selectively, the model can overfit and experience sustained performance degradation after the drift subsides.

Therefore, a proactive online adaptation framework is needed to (i) enable rapid parameter adjustment under drift, and (ii) explicitly classify stable, long-term changes from fluctuations driven by variant unobserved factors, so that different online learning strategies can be applied accordingly.

C. Structural Causal Model

To provide a principled basis for such selective adaptation, Structural Causal Model (SCM) is introduced to represent latent drivers and causal mechanisms, providing a principled basis for reasoning. SCM consists of endogenous variables V, exogenous variables U, structural assignments $V_i \leftarrow f_i(\text{Pa}(V_i), U_i)$ for each $V_i \in V$, and a joint distribution $P(U)$ over the exogenous variables. The directed graph induced by the parent sets Pa_i encodes direct causal effects among variables. The induced directed graph encodes direct causal effects.

III. Methodology

A. Overview

RQ1:
RQ2:
RQ3:

The methodology is organized as follows: invariant learning for stable components is defined first, followed by the factor-mechanism forecasting backbone, and finally the two-stage online adaptation.

B. The decomposition of ... using Invariant Learning

To separate long-term changes from variant fluctuations, an invariant learning module is introduced to extract features $\mathbf{X}_{\text{I}}$ and a sufficiently predictable component $\mathbf{Y}_{\text{suf}}$ whose conditional relation $P_e(\mathbf{Y}_{\text{suf}}|\mathbf{X}_{\text{I}})$ is invariant across environments (Fig. 1(b)). $\mathbf{X}_{\text{I}}$ is the subset of inputs $\mathbf{X}$ that provides sufficient, environment-stable predictive information for $\mathbf{Y}_{\text{suf}}$, and both are intended to exclude variant effects from unobserved factors. The following invariance assumption is adopted:

$$\mathbf{Y} = \alpha(Z)\mathbf{Y}_{\text{suf}} + \beta(Z)\mathbf{1}. \tag{1}$$

Here $\alpha(Z)$ and $\beta(Z)$ are mapping functions of Z. The effect of Z is mitigated via Instance Residual Normalization (IRN):

$$\tilde{\text{Res}}_t = \frac{Y_t - \mu(Y_t)}{\sigma(Y_t)} - \frac{\widehat{Y}_t - \mu(\widehat{Y}_t)}{\sigma(\widehat{Y}_t)}, \tag{2}$$

where $\mu(\cdot)$ and $\sigma(\cdot)$ denote the mean and standard deviation. IRN ensures the residuals have mean 0 and variance $2-2\text{cov}(\widehat{Y}, Y)$, where cov denotes the covariance. This yields the surrogate loss

$$\ell(\widehat{Y}, Y) = \text{MSE}(\tilde{\text{Res}}, 0). \tag{3}$$

To mitigate the effect of Z on X, an adversarial pair $\mathcal{M}_{\text{EI}}$ and $\mathcal{M}_{\text{IL}}$ is optimized in an alternating manner. $\mathcal{M}_{\text{EI}}$ infers environments from the variant features retained by $\mathcal{M}_{\text{IL}}$, while $\mathcal{M}_{\text{IL}}$ removes features that enable accurate environment inference. At convergence, $\mathcal{M}_{\text{IL}}$ preserves invariant features $\phi^*(X)$ and $\mathcal{M}_{\text{EI}}$ fails to predict environment labels E_{infer}.

Specifically, $\mathcal{M}_{\text{EI}}$ maintains multiple environment-specific regressors $\rho^{(e)}$; each regressor defines a prototype prediction that serves as the center of a cluster in the representation space. The distance to environment e is measured by the sufficient loss $\ell_{\text{suf}}(\rho^{(e)}(\phi^*(X_t)), Y_t)$, and labels are assigned to the closest center. Let D_e be the samples assigned to environment e. This phase consists of two substeps: regression fitting and label reassignment. In the regression fitting step, we fit $\rho^{(e)}$ by minimizing the regression loss on each environment:

$$\begin{aligned}&\mathbb{E}_{e\in E_{\text{infer}}} R^{(e)}(\rho, \phi) \\ &= \min_{\{\rho^{(e)}\}} \sum_{e\in E_{\text{infer}}} \frac{1}{|D_e|} \sum_{(X,Y)\in D_e} \ell\Big(\rho^{(e)}(\phi^*(X)), Y\Big).\end{aligned} \tag{4}$$

In the reassignment step, the environment label of each sample is updated by the shortest loss:

$$E_{\text{infer}}(t) \leftarrow \arg\min_{e\in E_{\text{infer}}} \ell\Big(\rho^{(e)}(\phi^*(X_t)), Y_t\Big), \tag{5}$$

and optionally apply a temporal smoothing step via majority voting over a radius r:

$$E_{\text{infer}}(t) \leftarrow \text{mode}\{E_{\text{infer}}(t+j)\}_{j=-r}^{r}. \tag{6}$$

The invariant-learning module consists of two branches: an invariant branch $\mathcal{M}_{\text{IL}}$ and a variant branch $\mathcal{M}_{\text{V}}$, with

output embeddings $\phi_{\mathrm{I}}(X_t)$ and $\phi_{\mathrm{V}}(X_t)$. The invariant branch is learned by minimizing the environment-robust objective:

$$\min_{\rho,\phi} \mathcal{L}_{\mathrm{TIL}} = \mathbb{E}_{e\in E_{\mathrm{infer}}} R^{(e)}(\rho,\phi) + \lambda_1 \operatorname{Var}_{e\in E_{\mathrm{infer}}} R^{(e)}_{\mathrm{suf}}(\rho,\phi), \tag{7}$$

where the variance penalty $\operatorname{Var}_{e\in E_{\mathrm{infer}}} R^{(e)}_{\mathrm{suf}}(\rho,\phi)$ encourages invariance across environments. The variant branch shares the same architecture and embedding dimensionality but relaxes the variance penalty via a smaller λ_2, retaining more variation. This design enables controlled comparison of how variant fluctuations influence embeddings and provides a basis for subsequent online adaptation strategies that explicitly target variant effects.

Implementation details of $\mathcal{M}_{\mathrm{EI}}$ and $\mathcal{M}_{\mathrm{IL}}$ are summarized in Section IV-E. Next, the factor-mechanism modeling that underpins the forecasting backbone and enables expert-wise causal analysis is described.

C. C

In real-world load forecasting, observed covariates $\mathbf{x}_t$ (e.g., temperature, humidity, wind speed) rarely cause the load $\mathbf{Y}_t$ directly. Instead, $\mathbf{x}_t$ is assumed to act through latent factor states $\mathbf{F}_t$, each representing an unobserved but physically meaningful driver (e.g., air-conditioning usage intensity or industrial activity). These factors are causally independent, respond to exogenous conditions $\mathbf{x}_t$, and contribute to $\mathbf{Y}_t$ in parallel. The goal is to disentangle factor-specific mechanisms and their time-varying contributions rather than attributing $\mathbf{Y}_t$ directly to $\mathbf{x}_t$.

Inspired by the conditional-mixture view of dynamic mechanisms, a dynamic graph is modeled as a conditionally weighted superposition of multiple static graphs. Each static graph encodes a distinct, potentially nonlinear interaction pattern, and the weights vary with system state, yielding a factor-model view of dynamic causality.

Consider a multivariate time series $\{\mathbf{x}_t\}_{t=1}^{T}$ with $\mathbf{x}_t \in \mathbb{R}^N$. We represent the target $\mathbf{Y}_t$ as a weighted combination of n_k factor generators:

$$\begin{aligned}\mathbf{Y}_t &= \boldsymbol{\alpha}(t) \odot^{\dagger} \mathbf{f}_{\phi}(\mathbf{x}_{t-L:t}) \\ &= \sum_{k=1}^{n_k} \alpha_k(t)\, f_{\phi_k}(\mathbf{x}_{t-L:t})\,,\end{aligned} \tag{8}$$

where f_{ϕ_k} is the k-th factor generator and $\alpha_k(t)$ is its coefficient during time window $t-L:t$. Factor coefficients $\boldsymbol{\alpha}(t)$ are obtained through a feature mapping of $\mathbf{z}_t$, and they are both assumed stable within the time window:

$$\begin{aligned}\mathbf{h}_t &= m_{\psi}(\mathbf{x}_{t-L:t})\,, \\ \mathbf{z}_t &= m_{\theta}(\mathbf{h}_t)\,, \\ \boldsymbol{\alpha}(t) &= m_{\eta}(\mathbf{z}_t) \in \mathbb{R}^{n_k},\end{aligned} \tag{9}$$

Here m_{ψ} is a temporal Transformer encoder with pooling that yields the summary embedding $\mathbf{h}_t$, m_{θ} is a variational encoder that models factor features $\mathbf{z}_t$ with mean and variance predicted from $\mathbf{h}_t$, and m_{η} maps $\mathbf{z}_t$ to coefficients $\boldsymbol{\alpha}(t)$. Substituting these components yields the forecasting form:

$$\mathbf{Y}_t = m_{\eta}\left(m_{\theta}\left(m_{\psi}(\mathbf{x}_{t-L:t})\right)\right) \odot^{\dagger} \mathbf{f}_{\phi}(\mathbf{x}_{t-L:t}) \tag{10}$$

We next detail the temporal encoder m_{ψ}, which produces timewise representations via standard self-attention blocks along the time axis. The input is first embedded as $\mathbf{H}_t^{(0)} = \mathbf{Embedding}(\mathbf{X}_t) \in \mathbb{R}^{N\times d_h}$. For each Transformer block, the computation is:

$$\begin{aligned}\mathbf{Q} = \mathbf{H}_t^{(l)}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{H}_t^{(l)}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{H}_t^{(l)}\mathbf{W}_V, \\ \operatorname{Attn}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \operatorname{softmax}\!\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_h}}\right)\mathbf{V}, \\ \mathbf{A}^{(l)} = \mathrm{MHA}(\mathbf{H}_t^{(l)})\mathbf{W}_O, \\ \widetilde{\mathbf{H}}_t^{(l)} = \mathrm{LN}\Big(\mathbf{H}_t^{(l)} + \mathbf{A}^{(l)}\Big), \\ \mathbf{H}_t^{(l+1)} = \mathrm{LN}\Big(\widetilde{\mathbf{H}}_t^{(l)} + \mathrm{FFN}\Big(\widetilde{\mathbf{H}}_t^{(l)}\Big)\Big).\end{aligned} \tag{11}$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d_h \times d_{(QKV)}}$, and the corresponding global summary embedding $\mathbf{h}_t \in \mathbb{R}^{d_{\mathrm{h}}}$ is obtained from the final layer output $\mathbf{H}_t^{(F)} \in \mathbb{R}^{N\times d_h}$ as $\mathbf{h}_t = \mathrm{MLP}_h\Big(\mathbf{H}_t^{(F)}\Big)$.

Given the window representation, a feature vector $\mathbf{z}_t \in \mathbb{R}^{n_k d_b}$ with factor blocks $\mathbf{z}_t = [\mathbf{z}_t^{(1)}; \ldots; \mathbf{z}_t^{(n_k)}]$ is introduced, where $\mathbf{z}_t^{(m)} \in \mathbb{R}^{d_b}$ denotes the latent feature associated with the m-th factor mechanism and d_b is a shared hyperparameter that sets the dimensions for all factor features. The feature vector $\mathbf{z}_t$ is inferred by a variational encoder with prior distribution $p(\mathbf{z}_t)$ defined as a standard Gaussian and variational posterior $q_{\zeta}(\mathbf{z}_t \mid \mathbf{X}_t)$ parameterized by encoder-predicted mean and diagonal variance:

$$\begin{aligned}\mathbf{z}_t &\sim \mathcal{N}(\mathbf{0},\mathbf{I}), \\ q_{\zeta}(\mathbf{z}_t \mid \mathbf{X}_t) &= \mathcal{N}\big(\boldsymbol{\mu}_{\zeta}(\mathbf{h}_t), \operatorname{diag}(\boldsymbol{\sigma}^2_{\zeta}(\mathbf{h}_t))\big).\end{aligned} \tag{12}$$

The decoder reconstructs the window via a feedforward network, $\widehat{\mathbf{X}}_t = \mathrm{Dec}_x(\mathbf{z}_t)$. We then map the latent features to factor weights, assumed stable within the window: $\boldsymbol{\alpha}(t) = m_{\eta}(\mathbf{z}_t) \in \mathbb{R}^{n_k}$.

For factor-wise prediction, a sparse mixture-of-experts (MoE) separates the modeling of distinct factors. Each expert corresponds to f_{ϕ_k} and is implemented as a Transformer block dedicated to the k-th factor; unlike the temporal encoder, these blocks employ multivariate self-attention along the variable axis and yield embeddings $\mathbf{e}_{t,\ell}^{(m)}$. The expert embeddings are first fused with $\boldsymbol{\alpha}(t)$ and then mapped to the output:

$$\mathbf{e}_{t,\ell} = \sum_{k=1}^{n_k} \alpha_k(t)\, \mathbf{e}_{t,\ell}^{(m)}, \qquad \widehat{\mathbf{y}}_{t,\ell} = f_o(\mathbf{e}_{t,\ell}). \tag{13}$$

To ensure that experts encode distinct and stable mechanisms rather than redundant correlations, we introduce a relevance-based regularizer. Following layer-wise relevance propagation (LRP), the prediction is decomposed into node-level relevance scores that are propagated backward through the network. Specifically,

$$\sum_{d\in V_L} R_d^{(L)} = \cdots = \sum_{d\in V_l} R_d^{(l)} = \cdots = \sum_{d\in V_1} R_d^{(1)}. \tag{14}$$

where V_l denotes the set of nodes in layer l, and $R_d^{(l)}$ denotes the relevance score of node $d \in V_l$, namely, its contribution to the final prediction at layer l. This equality explicitly describes the cross-layer contribution relation: the total relevance is

conserved when propagated from deep to shallow layers. For the transition from layer $l+1$ to l, relevance is redistributed as:

$$R_i^{(l)} = \sum_{j \in V_{l+1}} \frac{a_i^{(l)} w_{ij}}{\sum_{i' \in V_l} a_{i'}^{(l)} w_{i'j} + b_j} R_j^{(l+1)}, \tag{15}$$

which further implies the local conservation relation $\sum_{i \in V_l} R_{i \leftarrow j}^{(l)} = R_j^{(l+1)}, \forall\, j \in V_{l+1}$. Here, $a_i^{(l)}$ is the activation of node i in layer l, w_{ij} is the weight from node i to node j, and b_j is the bias term of node j. Larger positive relevance indicates stronger causal contribution.

To quantify differences in the mechanisms captured by factor experts, we define an expert-wise discrepancy using relevance maps. For a target variable i, each expert yields a two-dimensional LRP relevance map over the window $t\text{–}L : t$, denoted by $\mathbf{R}_{i,t}^{(m)} \in \mathbb{R}^{N \times L}$, $m = 1, \ldots, n_k$, with entries:

$$\mathbf{R}_{i,t}^{(m)}[v, \tau] = \left| R_{d(v,\tau)}^{(1, m \to i, t)} \right|, \quad v = 1, \ldots, N,\ \tau = 1, \ldots, L. \tag{16}$$

where $R_{d(v,\tau)}^{(1,m \to i,t)}$ is the input-layer relevance for variable v at lag τ when expert m explains target i at time t, and $|\cdot|$ keeps magnitude. We treat $\mathbf{R}_{i,t}^{(m)}$ as an $N \times L$ relevance surface and compute discrepancy per variable along the temporal axis. For variable $v \in \{1, \ldots, N\}$, let $\mathbf{r}_{i,v,t}^{(m)} = \mathbf{R}_{i,t}^{(m)}[v, :] \in \mathbb{R}^L$. To jointly capture temporal-shape mismatch and contribution-magnitude mismatch, a composite discrepancy is defined. First, to quantify contribution magnitude, define the total positive mass:

$$M_{i,v}^{(m)}(t) = \sum_{\tau=1}^{L} \mathbf{r}_{i,v,t}^{(m)}(\tau), \tag{17}$$

Then, to quantify temporal-shape mismatch, each profile $\mathbf{r}_{i,v,t}^{(m)}$ is normalized into a lag distribution:

$$\tilde{\mathbf{p}}_{i,v,t}^{(m)}(\tau) = \frac{\mathbf{r}_{i,v,t}^{(m)}(\tau)}{M_{i,v}^{(m)}(t) + \delta}, \quad \tau = 1, \ldots, L. \tag{18}$$

Here, $\tilde{\mathbf{p}}_{i,v,t}^{(m)}$ is the lag distribution of expert m for variable v when explaining target i at time t. For two experts indexed by p and q, the shape discrepancy on variable v is:

$$\Delta_{i,v}^{\text{shape}}(p, q; t) = W_1\left(\tilde{\mathbf{p}}_{i,v,t}^{(p)}, \tilde{\mathbf{p}}_{i,v,t}^{(q)}\right), \tag{19}$$

where $W_1(\cdot, \cdot)$ denotes the Wasserstein Distance between two discrete lag distributions. In parallel, the magnitude discrepancy is defined as:

$$\Delta_{i,v}^{\text{mass}}(p, q; t) = \left| M_{i,v}^{(p)}(t) - M_{i,v}^{(q)}(t) \right|. \tag{20}$$

Accordingly, the variable-level discrepancy $\Delta_{i,v}^{(p,q)}$ is defined by combining the shape and magnitude terms:

$$\Delta_{i,v}^{(p,q)}(t) = \Delta_{i,v}^{\text{shape}}(p, q; t) + \lambda_{\text{mass}} \Delta_{i,v}^{\text{mass}}(p, q; t), \tag{21}$$

where λ_{mass} is the hyperparameter that balances shape mismatch and magnitude mismatch. Finally, the overall discrepancy $\mathcal{D}_i(t)$ is obtained by aggregating over variables and expert pairs:

$$\mathcal{D}_i(t) = \sum_{v=1}^{N} \sum_{1 \le p < q \le n_k} \Delta_{i,v}^{(p,q)}(t). \tag{22}$$

The overall objective function $\mathcal{L}(t)$ consists of the supervised prediction loss, the VAE reconstruction (ELBO) loss, and the relevance-based regularizer. Specifically,

$$\mathcal{L}(t) = \mathcal{L}_{\text{pred}}(t) + \lambda_1 \mathcal{L}_{\text{VAE}}(t) + \lambda_2 \frac{1}{N} \sum_{i=1}^{N} \mathcal{D}_i(t), \tag{23}$$

where λ_1 and λ_2 weight the VAE loss and the relevance-based regularizer, respectively. The supervised prediction loss over the window is:

$$\mathcal{L}_{\text{pred}}(t) = \sum_{\ell=1}^{L} \ell(\mathbf{y}_{t-L+\ell}, \widehat{\mathbf{y}}_{t,\ell}). \tag{24}$$

The VAE objective is composed of a reconstruction term given by the negative log-likelihood and a KL divergence regularizer:

$$\begin{aligned} \mathcal{L}_{\text{VAE}}(t) &= \mathbb{E}_{q_\zeta}[-\log p_\theta(\mathbf{X}_t \mid \mathbf{z}_t)] \\ &+ \beta D_{\text{KL}}\big(\mathcal{N}(\boldsymbol{\mu}_\zeta(\mathbf{h}_t), \text{diag}(\boldsymbol{\sigma}_\zeta^2(\mathbf{h}_t))) \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{I})\big). \end{aligned} \tag{25}$$

D. D

We now present the two-stage online adaptation that operates on top of the factor-mechanism backbone. **Stage I (weight-only adaptation under stable mechanisms).** We introduce a stability metric to quantify whether each expert's causal mechanism remains consistent over short horizons. Using the previously defined LRP relevance map $\mathbf{R}_{i,t}^{(m)}$, we normalize it as $\hat{\mathbf{R}}_{i,t}^{(m)} = \mathbf{R}_{i,t}^{(m)} / \|\mathbf{R}_{i,t}^{(m)}\|_1$ and define a temporal stability score

$$S_i^{(m)}(t) = \frac{1}{|\mathcal{W}|} \sum_{\tau \in \mathcal{W}} \cos\left(\hat{\mathbf{R}}_{i,t}^{(m)}, \hat{\mathbf{R}}_{i,t-\tau}^{(m)}\right), \tag{26}$$

where $\mathcal{W}$ is a short rolling window. This score measures the consistency of relevance distributions over time. An expert is considered stable if $S_i^{(m)}(t) \ge \gamma$, and Stage I is activated for stable experts.

With stable mechanisms identified, we focus on the remaining short-term variability in how factor features are combined. In this stage, each expert's internal parameters are kept fixed and online adaptation updates only the combination weights. As defined earlier, the factor coefficients are produced from the VAE latent feature vector $\mathbf{z}_t$ through a lightweight feed-forward network m_η, yielding dynamic weights $\boldsymbol{\alpha}(t) = m_\eta(\mathbf{z}_t)$. The design supports proactive adaptation before labels arrive because the combination weights are generated in real time from $\mathbf{z}_t$. By default, the parameters of the VAE and each expert remain frozen and only m_η is updated online when delayed labels become available to better recover factor weights from $\mathbf{z}_t$.

Stage II (mechanism-aware adaptation for unstable experts). LRP relevance maps are jointly influenced by the input sample $\mathbf{X}_t$, the model parameters θ, and the factor mechanisms instantiated by each expert. Hence, attributing relevance drift solely to mechanism change is confounded by both parameter evolution and input distribution shift. Stage II addresses this by (i) stabilizing relevance maps with a slow-moving reference model and (ii) aligning LRP drift to parameter updates via a low-rank adaptation generator while restricting updates to unstable experts and their LRP-localized parameter subsets.

We maintain a reference model with the same architecture but slower weight dynamics, updated by an exponential moving average every K optimization steps:

$$\theta_{\text{ref}} \leftarrow (1-\eta)\theta_{\text{ref}} + \eta\,\theta, \qquad \eta \in [0,1). \tag{27}$$

A scheduled η balances early adaptation and long-term stability, allowing θ_{ref} to provide stable relevance signals while tracking gradual mechanism evolution.

For the same input window $\mathbf{X}_t$, we compute LRP maps $\mathbf{R}_{i,t}^{(m)}(\theta)$ and $\mathbf{R}_{i,t}^{(m)}(\theta_{\text{ref}})$ from the current and reference models. Their difference isolates mechanism/parameter discrepancies while suppressing confounding by input shift:

$$\Delta\mathbf{R}_{i,t}^{(m)} = \mathbf{R}_{i,t}^{(m)}(\theta) - \mathbf{R}_{i,t}^{(m)}(\theta_{\text{ref}}). \tag{28}$$

Let the unstable expert set be $\mathcal{U}_t = \{m \mid S_i^{(m)}(t) < \gamma\}$, where $S_i^{(m)}(t)$ is the stability score from Stage I. Stage II updates only the parameters associated with experts in $\mathcal{U}_t$.

For each unstable expert, we first identify input-layer relevance locations with the largest discrepancies, then back-trace these locations to locate the most influential neurons in each layer. The input-layer relevance map is indexed by variable–lag pairs (v,τ); each location corresponds to node $d(v,\tau)$. Let $\Omega_t = \text{TopK}(|\Delta\mathbf{R}_t^{(1)}|, k_\Omega)$ denote the set of variable–lag locations with the largest discrepancy. We propagate relevance from Ω_t to each layer and select the neurons that contribute most to those locations. At layer l, we define:

$$\mathcal{S}_t^{(l)} = \text{TopK}\Big(|\Delta\mathbf{R}_{t,\Omega}^{(l)}|, k\Big), \qquad \mathcal{S}_t = \bigcup_l \mathcal{S}_t^{(l)}. \tag{29}$$

Only parameters in $\mathcal{S}_t$ (restricted to experts in $\mathcal{U}_t$) are adapted.

We embed the relevance maps using lightweight LRP encoders:

$$\mathbf{h}_t = \text{Enc}_{\text{LRP}}\Big(\mathbf{R}_{i,t}^{(m)}(\theta)\Big), \quad \mathbf{h}_t^{\text{ref}} = \text{Enc}_{\text{LRP}}\Big(\mathbf{R}_{i,t}^{(m)}(\theta_{\text{ref}})\Big), \tag{30}$$

and define the drift representation $\mathbf{d}_t = \mathbf{h}_t - \mathbf{h}_t^{\text{ref}}$.

To align LRP drift with parameter changes, we use a parameter-efficient adaptation generator. For simplicity, consider a linear transformation in the l-th layer with parameters $\boldsymbol{\theta}^{(l)} \in \mathbb{R}^{d_{\text{in}}\times d_{\text{out}}}$. Given a small bottleneck $r \ll \min(d_{\text{in}}, d_{\text{out}})$, we compute adaptation coefficients by

$$\Big[\boldsymbol{\alpha}_t^{(l)}, \boldsymbol{\beta}_t^{(l)}\Big] = \mathbf{W}_2^{(l)}\,\sigma\Big(\mathbf{W}_1^{(l)}\,\mathbf{d}_t + \mathbf{b}^{(l)}\Big) + \mathbf{1}, \tag{31}$$

where $\mathbf{W}_1^{(l)} \in \mathbb{R}^{r\times d_c}$, $\mathbf{W}_2^{(l)} \in \mathbb{R}^{(d_{\text{in}}+d_{\text{out}})\times r}$, $\mathbf{b}^{(l)} \in \mathbb{R}^r$, d_c is the drift-embedding dimension, and $\sigma(\cdot)$ is the sigmoid. We use layer-wise bias terms $\mathbf{b}^{(l)}$ while sharing $\mathbf{W}_1$ and $\mathbf{W}_2$ across layers of the same type to reduce overhead.

We then construct a binary mask $\mathbf{M}_t^{(l)} \in \{0,1\}^{d_{\text{in}}\times d_{\text{out}}}$ from $\mathcal{S}_t^{(l)}$ and allow updates only on those entries. The adapted parameters are

$$\widehat{\boldsymbol{\theta}}_t^{(l)} = \Big(\mathbf{M}_t^{(l)} \odot \boldsymbol{\alpha}_t^{(l)}\boldsymbol{\beta}_t^{(l)\top}\Big) \odot \boldsymbol{\theta}_t^{(l)}. \tag{32}$$

Equivalently, the parameter adjustment is $\Delta\boldsymbol{\theta}_t^{(l)} = \widehat{\boldsymbol{\theta}}_t^{(l)} - \boldsymbol{\theta}_t^{(l)}$, which modifies only masked entries. This ensures that Stage II adapts only LRP-localized parameters of unstable experts, improving stability in online updates.

Stage III. To further enhance adaptation to variant fluctuations, a parallel mechanism is introduced that operates on the embeddings produced by the invariant and variant branches of the invariant-learning module. The same parameter-efficient generator is applied to both embeddings, and the resulting parameter shifts are optimized using the post-update prediction error.

Using the invariant and variant embeddings $\phi_{\text{I}}(X_t)$ and $\phi_{\text{V}}(X_t)$ at time t, the concept drift between the two branches is computed as their difference:

$$\mathbf{d}_t^{\text{V}} = \phi_{\text{V}}(X_t) - \phi_{\text{I}}(X_t). \tag{33}$$

This definition aligns with the role of the variant branch stated earlier: it preserves variation relative to the invariant branch and provides a reference for quantifying variant effects in the embedding space. The subsequent parameter update follows the same parameter-efficient adaptation generator as in Stage II, but uses $\mathbf{d}_t^{\text{V}}$ to target variant-induced deviations.

For a linear layer $\boldsymbol{\theta}^{(l)} \in \mathbb{R}^{d_{\text{in}}\times d_{\text{out}}}$, the shared low-rank generator produces adaptation coefficients from each drift,

$$\Big[\tilde{\boldsymbol{\alpha}}_t^{(l)}, \tilde{\boldsymbol{\beta}}_t^{(l)}\Big] = \tilde{\mathbf{W}}_2^{(l)}\,\sigma\Big(\tilde{\mathbf{W}}_1^{(l)}\,\mathbf{d}_t^{\text{V}} + \tilde{\mathbf{b}}^{(l)}\Big) + \mathbf{1}, \tag{34}$$

and the adapted parameters used for forecasting at time t are

$$\tilde{\boldsymbol{\theta}}_t^{(l)} = \Big(\tilde{\boldsymbol{\alpha}}_t^{(l)}\tilde{\boldsymbol{\beta}}_t^{(l)\top}\Big) \odot \widehat{\boldsymbol{\theta}}_t^{(l)}, \tag{35}$$

where $\odot$ denotes the element-wise product. The induced adjustment is

$$\Delta\boldsymbol{\theta}_t^{(l)} = \Big(\tilde{\boldsymbol{\alpha}}_t^{(l)}\tilde{\boldsymbol{\beta}}_t^{(l)\top} - \mathbf{1}\Big) \odot \widehat{\boldsymbol{\theta}}_t^{(l)}. \tag{36}$$

The generator parameters are then updated using the prediction error of the adapted model, which strengthens its ability to proactively adjust parameters under variant fluctuations.

Here $\widehat{\boldsymbol{\theta}}_t^{(l)}$ denotes the parameters after Stage II at time t. Stage III produces a temporary update on top of these Stage II parameters for the current step only. The fluctuation update $\Delta\boldsymbol{\theta}_t^{(l)} = \tilde{\boldsymbol{\theta}}_t^{(l)} - \widehat{\boldsymbol{\theta}}_t^{(l)}$ is applied only for prediction at time t and is discarded afterward. At time $t+1$, a new fluctuation update is computed on top of the updated Stage II parameters, so transient adaptations do not accumulate across steps.

E. E

We evaluate the proposed framework under both offline forecasting and online adaptation settings. The empirical study is organized around three questions: 1) whether the proposed factor-mechanism architecture provides a strong static forecasting backbone; 2) whether each online stage improves over a frozen deployment baseline under delayed feedback; and 3) whether combining Stage I, Stage II, and Stage III yields the strongest overall online performance.

1) Datasets: To align the experiments with the regional load forecasting objective of this paper, the main benchmarks are three multivariate load datasets with exogenous covariates: *Panama*, *Tetouan*, and *Spain Energy–Weather*. The *Panama* dataset contains the national electricity demand together with meteorological variables from multiple cities and calendar

indicators. The *Tetouan* dataset contains zonal power consumption together with local weather and solar-flow variables. The *Spain Energy–Weather* dataset contains hourly system load together with weather and degree-day features. In our experiments, the prediction targets are `nat_demand` for *Panama*, `Zone 1 Power Consumption` for *Tetouan*, and `actual_load` for *Spain Energy–Weather*. All datasets are processed in chronological order. No random shuffling is applied across the temporal axis so as to preserve causal ordering and maintain consistency with the online adaptation scenario.

In addition to these public benchmarks, we further curate a new *Lishui-9* load dataset family from operational records of nine county-level regions in Lishui, Zhejiang Province. The family contains nine county-level multivariate load forecasting tasks with matched meteorological covariates and serves as an additional real-world validation set for all key experimental modules.

For each dataset, the samples are first split chronologically into training and validation subsets with ratios of 55% and 10%, respectively. The remaining suffix is reserved for online evaluation under a shifted-stream protocol described later. This preserves chronological causality while ensuring that the online deployment stream differs from the offline training regime.

2) Data Processing: Given a multivariate time series with N variables, sliding-window samples are constructed with a lookback length $L = 24$ and a forecasting horizon $H = 12$. For each timestamp t, the model takes the historical window $\mathbf{X}_{t-L+1:t} \in \mathbb{R}^{L\times N}$ as input and predicts the target trajectory over the next H steps. All variables are standardized using only the statistics computed from the training split, and the same normalization parameters are then applied to the validation and test splits to avoid information leakage.

Standard cleaning and interpolation are applied when needed before sliding-window construction.

3) Implementation Details: Table I summarizes the default implementation settings used in the main experiments. The factor-mechanism backbone is implemented as a plug-and-play forecasting framework rather than a fixed predictor. Besides the vanilla Transformer, we instantiate the same framework with seven additional transformer-based forecasters, namely iTransformer, PatchTST, Informer, Autoformer, FEDformer, Pyraformer, and Crossformer, while keeping the invariant-learning module, factor-gating module, expert fusion module, and the three online adaptation stages unchanged.

For the broad backbone screening, all candidate backbones are evaluated under the same data split, normalization pipeline, and online protocol, so that the comparison isolates the effect of the forecasting backbone itself. The online-stage definitions are given once in the following protocol subsection.

4) Online Evaluation Protocol: The online stage is evaluated on the chronologically ordered test stream. Labels arrive with delayed feedback, and therefore parameter updates are only triggered after the corresponding delayed target becomes available. We compare the following variants:

- **Baseline**: the offline model is frozen and directly applied to the online stream;
- **Stage I**: only the factor-weight predictor is adapted online;
- **Stage II**: only the relevance-guided sparse parameter update is activated for unstable experts;
- **Stage III**: only the transient variant-drift compensation is activated during inference;
- **Full Model**: Stage I, Stage II, and Stage III are jointly enabled.

Stage I uses delayed-label supervision together with conservative bias updates, clipping, and rollback protection. Stage II is triggered only when the relevance-map stability score indicates that an expert becomes unstable, and only a small masked subset of expert parameters is allowed to change. Stage III computes a temporary correction from $\phi_V - \phi_I$ and applies it only to the current prediction step without permanently modifying the model parameters.

5) Evaluation Metrics: For static forecasting, we report Mean Squared Error (MSE) and Mean Absolute Error (MAE) on the validation and test sets. For online adaptation, we further record the real-time prediction error on the test stream and compare each adaptation variant against the frozen baseline. The main criterion in the online setting is whether each individual stage improves over the baseline and whether the joint Stage I+II+III configuration achieves the best overall online forecasting performance.

6) Distribution-Shifted Online Protocol: To better reflect the motivation of online learning, we additionally evaluate the strongest backbone under a more challenging shifted-stream protocol. Specifically, after fixing the offline training and validation segments, we search within the later candidate region for a contiguous online test block that maximizes the distribution-shift score relative to the training windows. The resulting shift scores are 0.1353 on *Panama*, 0.5036 on *Tetouan*, and 0.5504 on *Spain Energy–Weather*, indicating that the online streams used in this protocol differ substantially from the offline training distribution, especially on the latter two datasets.

7) Comparison with Online Forecasting Baselines: We evaluate against seven online baselines under the same shifted-stream protocol and the same iTransformer backbone. All methods start from the same frozen offline checkpoint; only the online update rule differs.

The first three are *model-agnostic online baselines*:

- **Frozen**: directly deploys the offline model without online adaptation.
- **GD**: performs full-parameter online gradient descent.
- **Head-GD**: restricts online gradient descent to the prediction head.

The remaining four are *representative online forecasting baselines*:

- **OneNet**: adapts a single forecasting network online.
- **SOLID**: uses self-supervised online adaptation under drift.
- **SOLID++**: strengthens SOLID with a more aggressive online controller.
- **PROCEED**: performs proactive drift-aware online adaptation.

TABLE I
IMPLEMENTATION DETAILS USED IN THE MAIN LOAD-FORECASTING EXPERIMENTS.

Category	Setting
Datasets	Panama, Tetouan, Spain Energy–Weather
Chronological split	Train/validation/online-test = 55%/10%/35%
Window setting	Lookback $L = 24$, forecast horizon $H = 12$
Normalization	Training-split standardization + RevIN in the forecasting backbone
Candidate backbones	Transformer, iTransformer, PatchTST, Informer, Autoformer, FEDformer, Pyraformer, Crossformer
Default backbone width	Hidden dimension 128, 4 attention heads, 2 encoder layers, dropout 0.1
Mechanism layer	$n_k = 4$ experts
Factor generator	Latent dimension 64; VAE gate in the default model
Offline optimizer	Adam with warmup-cosine schedule
Offline hyperparameters	Batch size 64, initial learning rate 5×10^{-4}, weight decay 5×10^{-5}, 5 epochs
Auxiliary loss weights	VAE weight 0.05, relevance discrepancy weight 0.05
Invariant learning	3 inferred environments, invariant dimension 64, smoothing radius 2, 1 pretraining epoch
Stage I	Delayed-label gate-bias update with clipping and rollback
Stage II	Sparse expert-only update with relevance-drift, reconstruction-error, and acceptance gates
Stage III	Bounded transient correction driven by $\phi_V - \phi_I$

TABLE II
COMPARISON WITH ONLINE FORECASTING BASELINES UNDER THE SHIFTED-STREAM PROTOCOL ON THE ITRANSFORMER BACKBONE. EACH DATASET HAS PAIRED MSE AND MAE SUBCOLUMNS. THE FIRST THREE ADAPTIVE BASELINES (**GD**, **HEAD-GD**, AND **ONENET**) ARE MODEL-AGNOSTIC, WHILE THE REMAINING METHODS ARE FORECASTING-TIME ADAPTATION BASELINES. THE *Lishui-9* COLUMNS REPORT THE FAMILY-LEVEL MEAN OVER THE NINE COUNTY TASKS. SHIFT SCORES ARE 0.1353 ON *Panama*, 0.5036 ON *Tetouan*, 0.5504 ON *Spain Energy–Weather*, AND 0.3027 ON *Lishui-9*. LOWER IS BETTER.

Method	Panama		Tetouan		Spain Energy–Weather		Lishui-9	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Frozen	0.2004	0.3041	0.0841	0.1931	0.2460	0.3576	0.2806	0.3990
GD	1.3342	0.8979	0.5094	0.4939	0.8037	0.6911	0.3750	0.4515
Head-GD	0.3634	0.4505	0.0943	0.2158	0.2859	0.3966	0.2804	0.3994
OneNet	0.3534	0.4416	0.0833	0.1932	0.2768	0.3882	0.2779	0.3976
SOLID	0.2019	0.3060	0.0893	0.2030	0.2518	0.3639	0.2762	0.3966
SOLID++	0.4500	0.4913	0.0940	0.2097	0.2992	0.4015	0.2757	0.3964
PROCEED	0.2042	0.3083	0.0933	0.2054	0.2420	0.3547	0.2795	0.3970
Ours-Stage III	0.1977	0.3005	0.0824	0.1907	0.2415	0.3540	0.2778	0.3972
Ours-Full	0.1974	0.3005	0.0820	0.1901	0.2417	0.3535	0.2775	0.3969

Table II summarizes the shifted-stream comparison under a shared iTransformer backbone and a shared frozen offline checkpoint. The first three adaptive baselines, **GD**, **Head-GD**, and **OneNet**, are model-agnostic online strategies; **SOLID**, **SOLID++**, and **PROCEED** are representative forecasting-time adaptation baselines. Each dataset is reported with paired MSE and MAE subcolumns. Fig. 13 further expands the *Lishui-9* family to the county level. Naive online gradient updates are unstable under delayed feedback and explicit distribution shift. In contrast, the proposed controller remains competitive across all four datasets: the full Stage I+II+III model performs best on *Panama* and *Tetouan*, Stage III alone performs best on *Spain Energy–Weather*, and both Ours-Stage III and Ours-Full consistently outperform **Frozen**, **GD**, **Head-GD**, **OneNet**, and **PROCEED** on the *Lishui-9* family, with the remaining gap concentrated against **SOLID** and **SOLID++**.

8) Broad Backbone Screening: Table III reports the broad plug-in screening over eight transformer-based backbones under a shared configuration and a shared online protocol. Three trends are immediately visible. First, the proposed framework still benefits from stronger temporal encoders, although the absolute error levels become higher than in the earlier pilot runs because the current benchmark uses a stricter shifted-stream protocol and the same controller settings for all candidate backbones. Under this harder setting, iTransformer remains the strongest backbone, while FEDformer and Crossformer form the second tier with a clear margin over the remaining candidates. Second, the Stage III transient correction yields consistent online gains across all screened backbones, indicating that the variant-drift compensation mechanism is not tied to any single predictor design. Third, the full Stage I+II+III pipeline remains competitive with Stage III and often further improves over the frozen baseline. This supports the central claim that the proposed method should be understood as a forecasting-and-adaptation framework rather than a customized wrapper around one specific transformer.

9) Online Variant Comparison Across Eight Backbones: The previous top-three comparison is expanded here to all eight candidate backbones so that the online behavior of Stage II and the full controller can be judged more comprehensively. Table IV reports the mean online MSE of the frozen baseline, Stage II, Stage III, and the full Stage I+II+III controller under the tuned online settings. Two conclusions are important. First, Stage III remains the most reliable single online component across all eight backbones. Second, Stage II is generally weaker than Stage III and often only yields modest gains. This does not indicate that Stage II is ineffective

TABLE III
BROAD BACKBONE SCREENING ON THE THREE MAIN LOAD-FORECASTING DATASETS. REPORTED NUMBERS ARE MEANS OVER *Panama*, *Tetouan*, AND *Spain Energy–Weather*. LOWER IS BETTER.

Backbone	Offline MSE	Offline MAE	Online Baseline	Online Stage III	Online Full
iTransformer	0.2071	0.3122	0.2077	0.2046	0.2047
FEDformer	0.2018	0.3165	0.2024	0.1995	0.1995
Crossformer	0.2445	0.3418	0.2442	0.2412	0.2412
Informer	0.3463	0.4073	0.3465	0.3440	0.3436
Autoformer	0.3723	0.4296	0.3728	0.3704	0.3708
Pyraformer	0.5549	0.5539	0.5554	0.5535	0.5536
Transformer	0.5772	0.5676	0.5781	0.5771	0.5770
PatchTST	0.6623	0.6200	0.6632	0.6626	0.6628

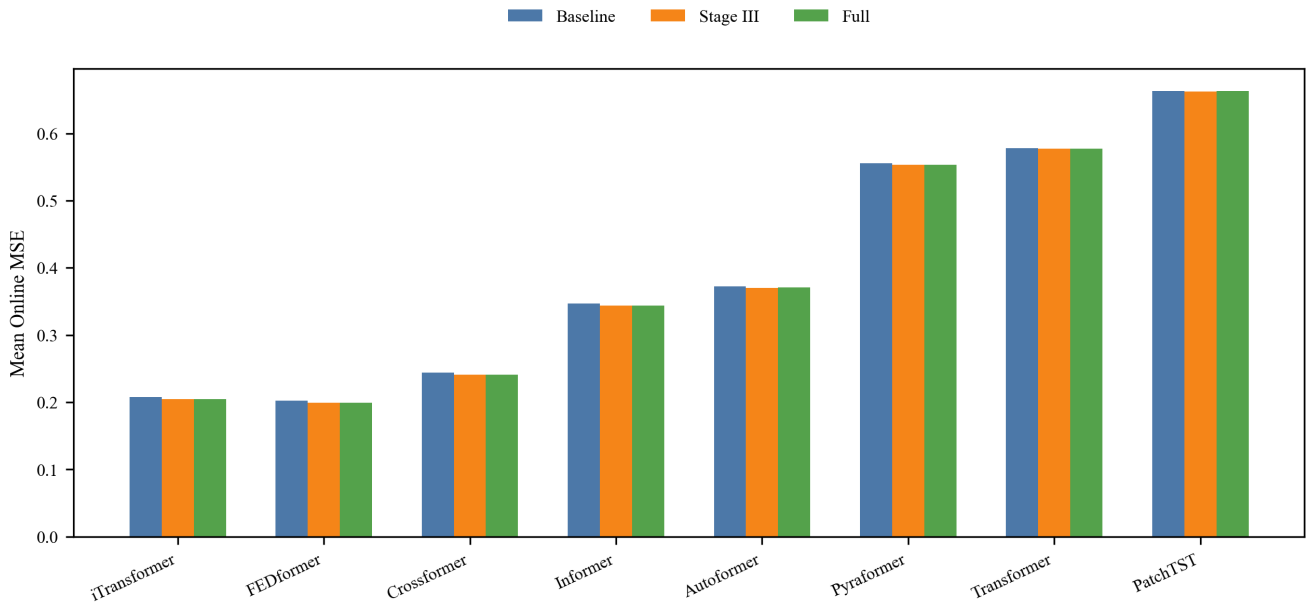


Fig. 2. Mean online MSE of the frozen baseline, Stage III, and the full Stage I+II+III pipeline across eight plug-in transformer backbones.

in principle; rather, Stage II depends on the quality of the relevance-drift estimate and on a sufficiently stable expert decomposition. When the backbone itself is weak, the input-layer discrepancy map is less informative, and the resulting sparse update becomes either noisy or over-conservative. Conversely, for stronger backbones such as iTransformer, FEDformer, and Crossformer, Stage II stays close to the frozen baseline and is most useful when combined with the other two stages, where Stage I stabilizes the factor mixture and Stage III absorbs the strongest short-term variation.

10) Ablation on Online Stage Combinations: To isolate the contribution of each online stage, we evaluate all seven non-empty stage combinations together with the frozen baseline under the same shifted-stream protocol. Instead of compressing the result into a public mean, Table V now reports the four datasets separately, and Fig. 4 visualizes them as four dataset-level panels. Fig. 14 further decomposes the Lishui family into nine county-level subplots. Three observations are important. First, Stage III remains the strongest single online module on all four datasets, which is consistent with the earlier finding that transient correction based on $\phi_V - \phi_I$ is the most reliable source of online gain. Second, Stage I alone provides only a small improvement, while Stage II alone remains neutral or slightly weaker on several datasets; this again suggests that the main contribution of Stage II is collaborative rather than standalone. Third, the full Stage I+II+III pipeline is strongest on *Spain Energy–Weather* and remains competitive on all the other datasets, while the Lishui county plots reveal that the relative ranking among Stage III, Stage II+III, and the full controller is county dependent. This confirms that Stage II becomes beneficial once it operates inside the complete controller, where Stage I stabilizes the factor mixture and Stage III handles the strongest short-term drift.

11) Component Ablation: Beyond the online-stage combinations, we further ablate the key structural components of the framework. The evaluated variants include removing the invariant-learning branch (**w/o Invariant**), disabling the discrepancy-localized sparse Stage II update and updating dense layer parameters instead (**Dense Stage II**), replacing the VAE gate by a simple MLP factor-weight generator (**MLP Gate, no VAE**), removing RevIN normalization (**w/o RevIN**), and collapsing the expert bank into a single expert (**Single Expert**). These are the common component variants for which we report all four datasets jointly. Controller-only ablations such as **w/o LRP in Stage II**, **w/o EMA Ref**, **w/o Rollback/Buffer**, and **Sync All Experts** remain discussed in the text and logs,

TABLE IV
ONLINE VARIANT COMPARISON ACROSS EIGHT TRANSFORMER-BASED BACKBONES UNDER THE TUNED CONTROLLER SETTINGS. NUMBERS ARE MEAN ONLINE MSE OVER THE THREE MAIN LOAD DATASETS. LOWER IS BETTER.

Backbone	Baseline	Stage II	Stage III	Stage I+II+III
iTransformer	0.2077	0.2078	0.2046	0.2047
FEDformer	0.2024	0.1892	0.1995	0.1995
Crossformer	0.2442	0.1965	0.2412	0.2412
Informer	0.3465	0.3048	0.3440	0.3436
Autoformer	0.3728	0.3472	0.3704	0.3708
Pyraformer	0.5554	0.5175	0.5535	0.5536
Transformer	0.5781	0.5631	0.5771	0.5770
PatchTST	0.6632	0.6510	0.6626	0.6628

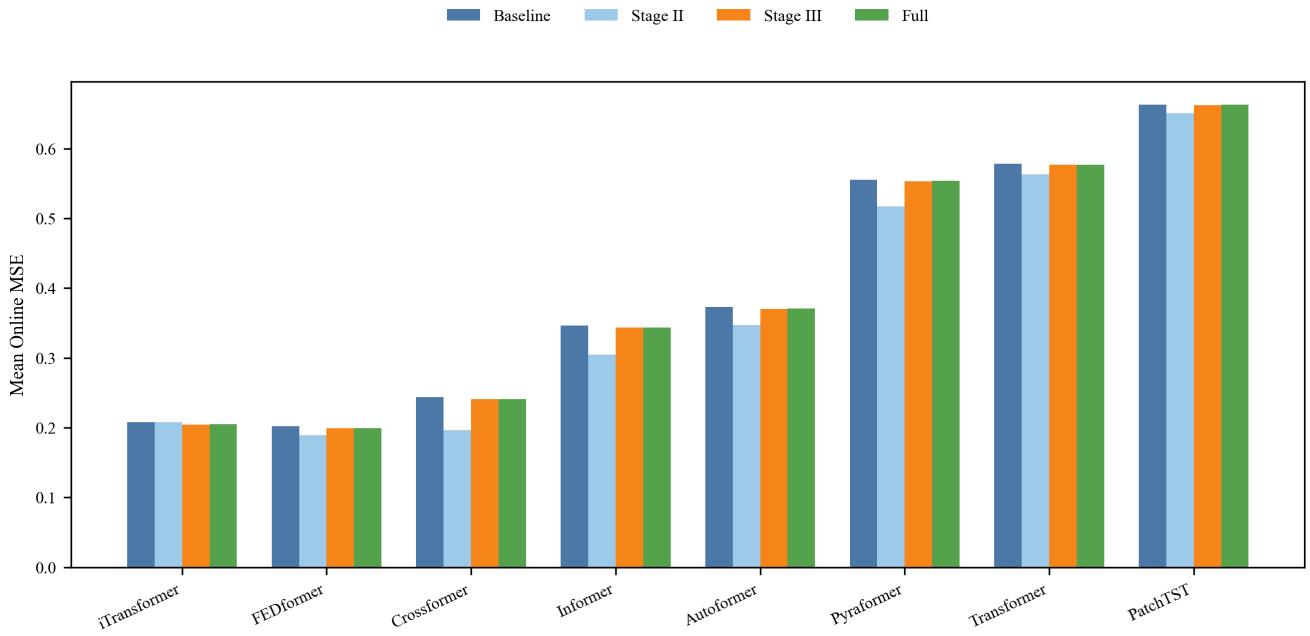


Fig. 3. Variant-level online comparison across eight transformer-based backbones under the tuned controller settings.

but are omitted from Table VI because they were not re-run for the full Lishui family under the same protocol. Table VI and Fig. 5 therefore focus on the shared ablations that are fully comparable across the four datasets, while Fig. 14 provides county-level detail for the Lishui family.

The component ablation reveals a clear hierarchy of importance. Removing RevIN or collapsing the multi-expert factor bank causes the largest degradation on both the public suite and the Lishui family, which shows that normalized input statistics and mechanism diversity are both critical for robust load forecasting under shift. Removing the invariant branch also degrades the result on most datasets, although the magnitude of the drop is smaller than for the normalization and expert-count ablations. Replacing the VAE gate by a simpler MLP gate remains competitive and even improves several datasets, including the Lishui family. This again suggests that the main gains come from the factorized expert framework and the online adaptation policy rather than from one single latent regularization form. We therefore treat the VAE gate as one valid instantiation of the framework rather than the only effective choice.

12) Case Study on Shifted Online Segments: To provide a more concrete view of the online behavior, we inspect one representative shifted segment from each of the four datasets. Earlier versions of the qualitative figure were visually dominated by the target and full-controller curves because the frozen, Stage III, and full predictions are often close on easy regions. Therefore, Fig. 6 now shows, for each dataset, an automatically selected informative window centered on the largest sustained frozen-versus-full disagreement. The left column plots the trajectories of the ground truth, the frozen baseline, Stage III, and the full Stage I+II+III controller, together with a zoomed inset on the most discriminative local region. The right column shows the corresponding absolute-error traces. This layout makes the comparison materially clearer: the frozen predictor usually captures the coarse load trend but exhibits visible lag and amplitude mismatch around abrupt local fluctuations, whereas Stage III and the full controller reduce these local deviations and lower the error envelope over the selected shifted windows.

13) Input Relevance Visualization: We also visualize the input-layer relevance map produced by the strongest online configuration on the same four datasets. Fig. 7 shows the

TABLE V
ABLATION ON ONLINE STAGE COMBINATIONS UNDER THE SHIFTED-STREAM PROTOCOL. THE FOUR COLUMNS REPORT *Panama*, *Tetouan*, *Spain Energy–Weather*, AND THE *Lishui-9* FAMILY MEAN, RESPECTIVELY. LOWER IS BETTER.

Variant	Panama	Tetouan	Spain Energy–Weather	Lishui-9
Baseline	0.2005	0.0835	0.2462	0.2271
Stage I	0.2007	0.0833	0.2451	0.2269
Stage II	0.2006	0.0837	0.2468	0.2271
Stage III	0.1979	0.0819	0.2427	**0.2254**
Stage I+II	0.2006	0.0839	0.2458	0.2269
Stage I+III	**0.1973**	0.0826	0.2424	0.2256
Stage II+III	0.1980	0.0826	0.2421	**0.2254**
Stage I+II+III	0.1974	**0.0821**	**0.2411**	0.2256

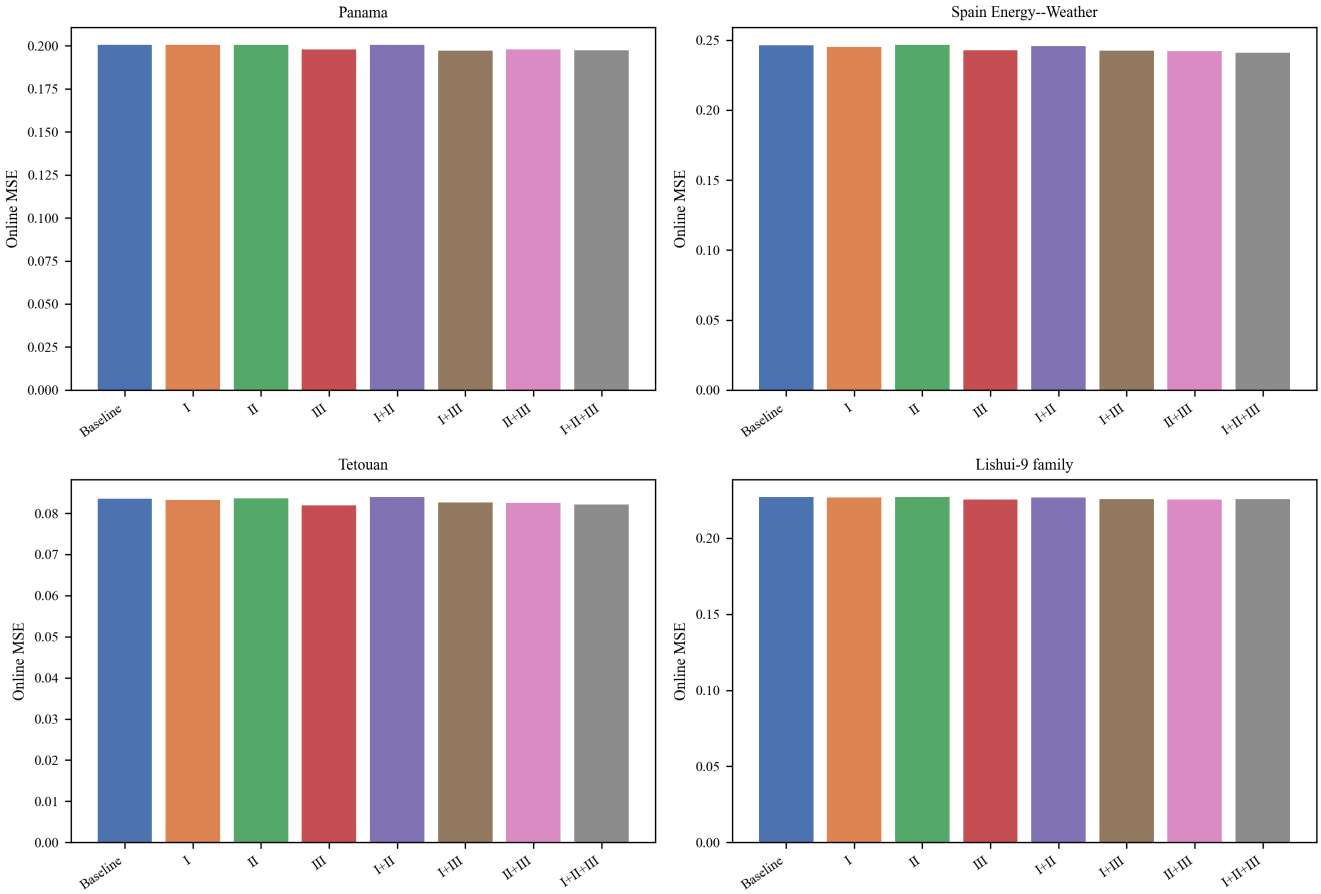


Fig. 4. Stage-combination ablation under the shifted-stream protocol. The four panels correspond to *Panama*, *Tetouan*, *Spain Energy–Weather*, and the *Lishui-9* family mean, respectively.

relevance heatmap of representative online steps selected from regions where the online correction produces one of the largest improvements over the frozen baseline. Across the four panels, the controller does not react uniformly over all covariates or lags. Instead, the dominant relevance mass is concentrated on a small subset of weather-sensitive and load-related variables and their recent lags. This pattern is physically plausible for regional load forecasting, where short-term demand corrections are often driven by temperature-sensitive consumption, precipitation-related behavioral changes, and the recent load state itself. The visualization therefore supports the interpretability claim of the proposed framework: the online controller is guided by a structured relevance surface rather than by an opaque global perturbation.

14) t-SNE Analysis of Representation and Controller States: Following the qualitative protocol used in recent online forecasting studies, we further visualize the learned representation spaces by t-SNE. For the three public datasets, Fig. 8 compares the invariant representation ϕ_I with the Stage III drift representation $\phi_V - \phi_I$ for train and online samples, Fig. 9 visualizes the Stage I gate space and the Stage II relevance-drift controller space, and Fig. 10 shows the Stage III gain space and the mechanism space induced by the expert branches. Because *Lishui-9* is a nine-task family rather than a single dataset, we visualize its counties separately in Figs. 16–17. Table VII summarizes the public-dataset indicators, while the

TABLE VI
COMMON COMPONENT ABLATION UNDER THE SHIFTED-STREAM PROTOCOL. THE FOUR COLUMNS REPORT *Panama*, *Tetouan*, *Spain Energy–Weather*, AND THE *Lishui-9* FAMILY MEAN, RESPECTIVELY. LOWER IS BETTER.

Variant	Panama	Tetouan	Spain Energy–Weather	Lishui-9
Full	0.1981	0.0821	0.2418	0.2777
w/o Invariant	0.2025	0.0852	0.2354	0.2816
Dense Stage II	0.1976	0.0822	0.2422	0.2777
MLP Gate, no VAE	**0.1813**	**0.0744**	**0.2408**	**0.2745**
w/o RevIN	0.2879	0.1515	0.1735	0.9454
Single Expert	0.8388	0.2351	0.8533	0.4309

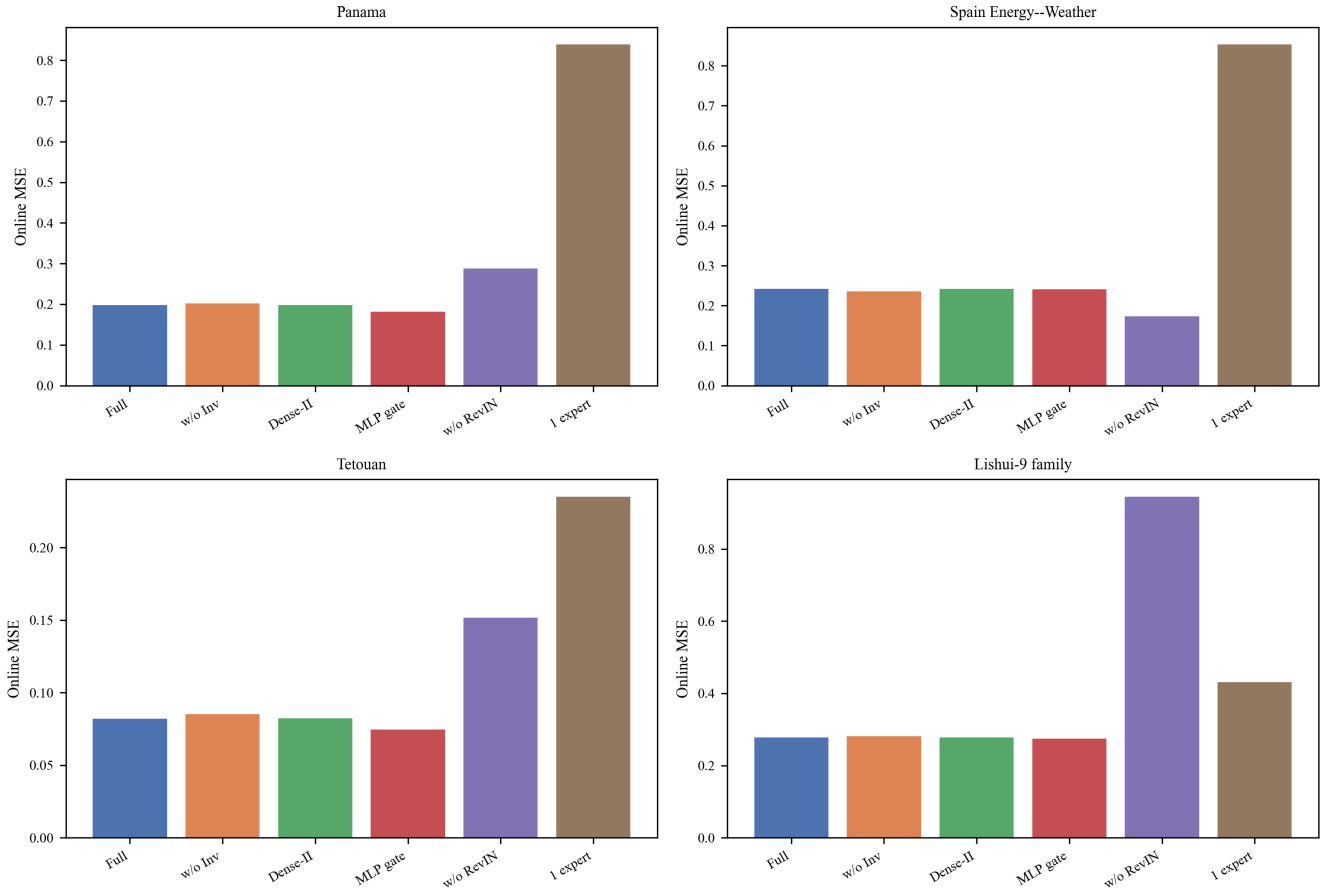


Fig. 5. Common component ablation under the shifted-stream protocol. The four panels correspond to *Panama*, *Tetouan*, *Spain Energy–Weather*, and the *Lishui-9* family mean, respectively.

Lishui family is analyzed visually at county resolution.

Three observations are consistent across the three datasets. First, the Stage III drift space is more sensitive to train–online mismatch than the invariant space. The centroid gap in $\phi_V - \phi_I$ is larger than that in ϕ_I on all datasets, e.g., $0.7536 > 0.5424$ on *Panama* and $1.3497 > 0.5288$ on *Tetouan*. This supports the intended role separation of the two branches: the invariant encoder preserves the stable forecasting structure, whereas the variant branch amplifies the distribution shift that should be handled online. Second, the Stage II controller remains highly sparse. The selected ratio is only 7.50% on *Panama*, 0.63% on *Tetouan*, and 8.75% on *Spain Energy–Weather*, which is consistent with the design of Stage II as a conservative correction mechanism rather than a dense online finetuning routine. Third, the expert mechanism space exhibits clear inter-expert separation, with mean pairwise centroid distances of 17.32, 13.64, and 17.58 on the three datasets, respectively. This indicates that the expert branches are not collapsing to a single shared solution, but are instead learning distinct factor-specific subspaces.

The Stage I gate-space visualization also provides a direct interpretation of the delayed online recalibration. The train–online gate gap is slightly reduced after Stage I is enabled on *Panama* and *Tetouan*, while it changes only marginally on *Spain Energy–Weather*. The Stage III scatter further shows that positive online gains are concentrated in localized drift regions rather than uniformly distributed across all online samples. Taken together, these plots indicate that Stage I mainly performs mild gate re-alignment, Stage II acts only on a small subset of samples with concentrated relevance-drift

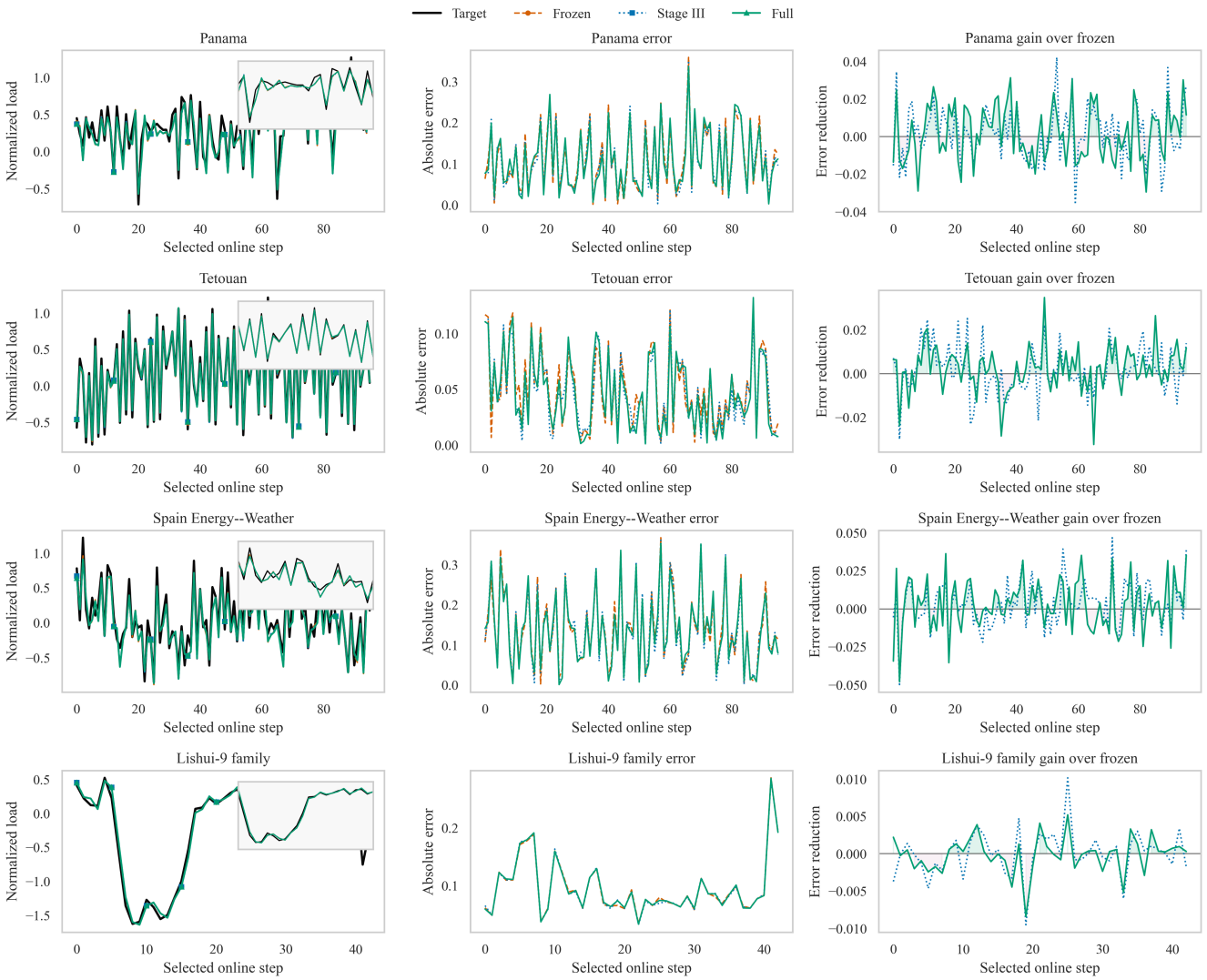


Fig. 6. Qualitative case studies on shifted online segments from the four datasets. For each dataset, the left subpanel shows the trajectories of the ground truth, the frozen baseline, Stage III, and the full Stage I+II+III controller over an automatically selected informative window, with a zoomed inset on the most discriminative local region. The right subpanel shows the corresponding absolute-error traces, which makes the differences between frozen, Stage III, and full adaptation visible even when the prediction curves are close.

TABLE VII

T-SNE SUMMARY ON THE THREE MAIN LOAD DATASETS. $\Delta\alpha$ DENOTES THE REDUCTION IN THE TRAIN–ONLINE GATE-SPACE CENTROID GAP AFTER STAGE I ADAPTATION. "SELECTED RATIO" IS THE PROPORTION OF ONLINE SAMPLES ACCEPTED BY THE STAGE II CONTROLLER. "STAGE III +GAIN" IS THE PROPORTION OF SAMPLED ONLINE POINTS ON WHICH STAGE III IMPROVES OVER THE FROZEN PREDICTION. "EXPERT MARGIN" IS THE MEAN PAIRWISE CENTROID DISTANCE AMONG THE EXPERT EMBEDDINGS.

Dataset	ϕ_I Gap	$(\phi_V - \phi_I)$ Gap	$\Delta\alpha$	Stage II Selected Ratio	Stage III +Gain	Expert Margin
Panama	0.5424	0.7536	0.0003	7.50%	51.88%	17.32
Tetouan	0.5288	1.3497	0.0012	0.63%	16.88%	13.64
Spain Energy–Weather	0.1619	0.2699	-0.0003	8.75%	46.25%	17.58

signatures, and Stage III exploits the more clearly shifted drift representation for transient correction.

15) Quantitative Effect of Shift Score: The previous low/medium/high grouping is useful for a first summary, but it is too coarse for judging online robustness. We therefore rewrite the shift-severity experiment as a quantitative shift-score analysis on each dataset. For every dataset, we sample multiple shifted online blocks from the post-validation region and report the exact shift score of each block, where the shift score is computed from the train–test covariate discrepancy. Table VIII reports the online MSE of the frozen model, Stage III, and the full controller on the resulting blocks, without averaging across datasets. Fig. 11 visualizes the same trend as four dataset-specific panels, where the fourth panel expands the Lishui family into county-level points rather than collapsing it into a single mean.

The resulting picture is more informative than a three-level average. On *Panama*, both Stage III and the full controller

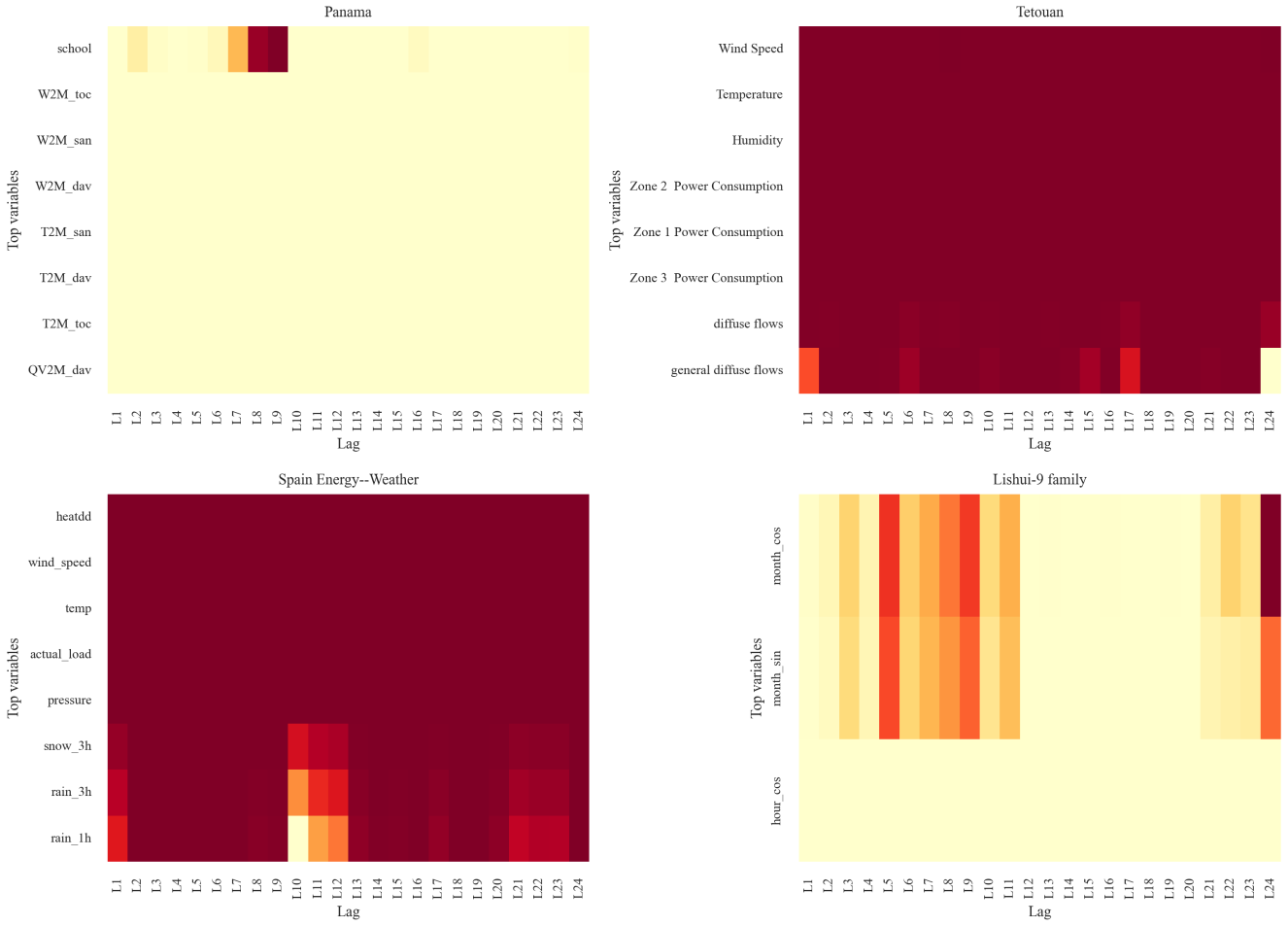


Fig. 7. Input-layer relevance heatmaps for representative online steps from the four datasets. The strongest relevance mass concentrates on a small subset of weather-sensitive and recent-load covariates, which is consistent with the operational semantics of short-term regional load forecasting.

remain below the frozen baseline throughout the tested shift-score range, with the full controller slightly better at the smallest sampled shift and Stage III slightly better at the larger two shift values. On *Tetouan*, the full controller performs best at the smallest shift score, whereas Stage III is marginally stronger at the largest shift. On *Spain Energy–Weather*, the full controller is strongest at all three sampled shift scores and remains consistently below the frozen baseline. Therefore, the main conclusion does not depend on a coarse severity label: the proposed online controller provides robust gains over a range of quantitatively measured train–test shifts, while the relative ranking between Stage III and the full controller is dataset dependent.

16) Additional Evaluation on the Newly Curated Lishui-9 Dataset Family: To verify that the framework is not tied to public benchmarks only, we additionally evaluate it on the newly curated *Lishui-9* dataset family, which contains nine county-level load forecasting tasks with aligned weather covariates. We first run the same family-level protocol as in the main experiments and average the resulting online error over the nine counties. Table IX shows that the qualitative trend remains consistent: Stage III and Stage II+III are the strongest family-level variants, while the full Stage I+II+III controller remains highly competitive. The gain is smaller than that on the three public datasets, which is expected because the county-level tasks are shorter and noisier after alignment, but the direction of the improvement remains stable.

We then perform a dedicated tuned shifted-stream comparison on the same family. Starting from the strongest Lishui-specific Crossformer checkpoint, we compare the frozen model, **SOLID++**, Ours-Stage III, and Ours-Full on the shifted online blocks of all nine counties. Table IX shows that the tuned version of the proposed controller narrows the gap to **SOLID++** substantially: the family-level mean online MSE decreases from 0.2265 for the frozen deployment to 0.2254 with Stage III and to 0.2255 with the full controller, while **SOLID++** reaches 0.2241. Although the strongest external baseline still remains slightly better on this newly curated family, the gap is reduced to approximately 1.3×10^{-3}, which is notably smaller than in the initial untuned Lishui runs. This additional result is important for two reasons: it confirms that the framework transfers to a newly curated regional load dataset with real alignment noise, and it shows that the main remaining challenge on this family is fine-grained online control rather than the static forecasting backbone itself.

Finally, the county-level views reveal substantial heterogeneity. Figs. 13–14 show that some counties benefit most from Stage III alone, whereas others benefit from Stage II+III

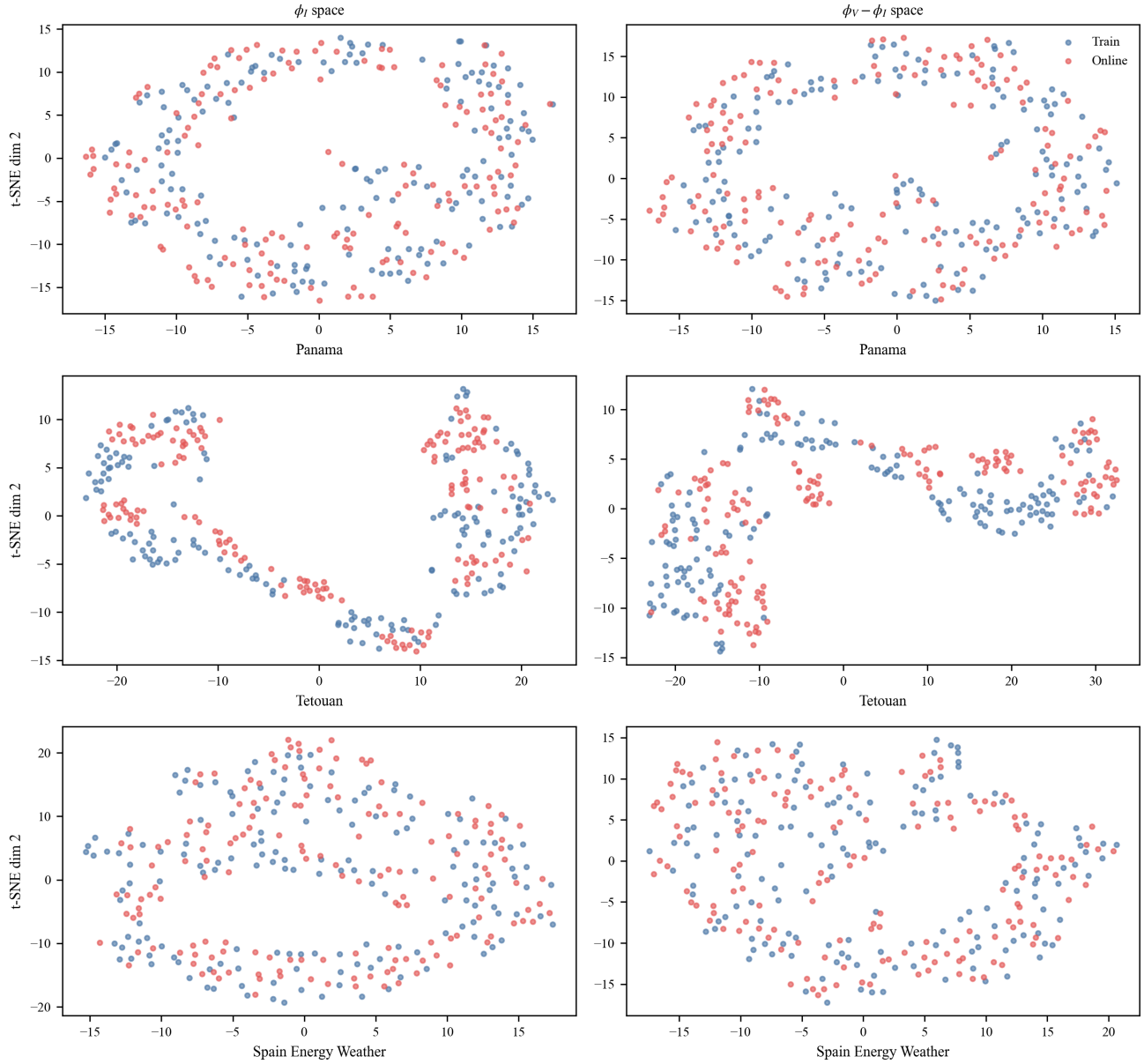


Fig. 8. t-SNE visualization of the invariant representation ϕ_I and the drift representation $\phi_V - \phi_I$ on the three main load datasets. The drift space exhibits a larger train–online separation than the invariant space, which is consistent with the intended decomposition between stable structure and online-varying dynamics.

or the full controller. The same phenomenon also appears in the county-level case studies and t-SNE plots in Figs. 15–17. Therefore, even on this newly curated dataset family, the core benefits still come from the normalization pipeline and the multi-expert factor mechanism, while the optimal online policy remains county dependent.

17) Analysis of Main Results: The current results support the two main claims of this paper. First, the proposed method is a *framework* rather than a model tied to one predictor. After replacing only the temporal encoder, the same invariant-learning module, factor-VAE module, expert mechanism layer, and online adaptation stages remain effective across a broad family of transformer-based forecasting backbones. Second, the online gains are not restricted to a single operating point. In the refined experiments, Stage III consistently improves the frozen baseline, while Stage I and Stage II provide additional corrections whose effect depends on the interaction between the backbone representation and the online controller.

At the dataset level, the strongest absolute online results are delivered by different backbones on different benchmarks. For iTransformer, the best results are 0.1802 on *Panama*, 0.0829 on *Tetouan*, and 0.2054 on *Spain Energy–Weather*. For FEDformer, the full combination reaches 0.1761 on *Panama*, while Stage III remains highly competitive on *Tetouan* and *Spain Energy–Weather*. For Crossformer, the full configuration achieves 0.2631 on *Spain Energy–Weather*, which provides the clearest example in the current study that the three online stages can deliver the strongest joint benefit on a tuned backbone. The newly curated *Lishui-9* family complements these public benchmarks: it confirms that the full controller

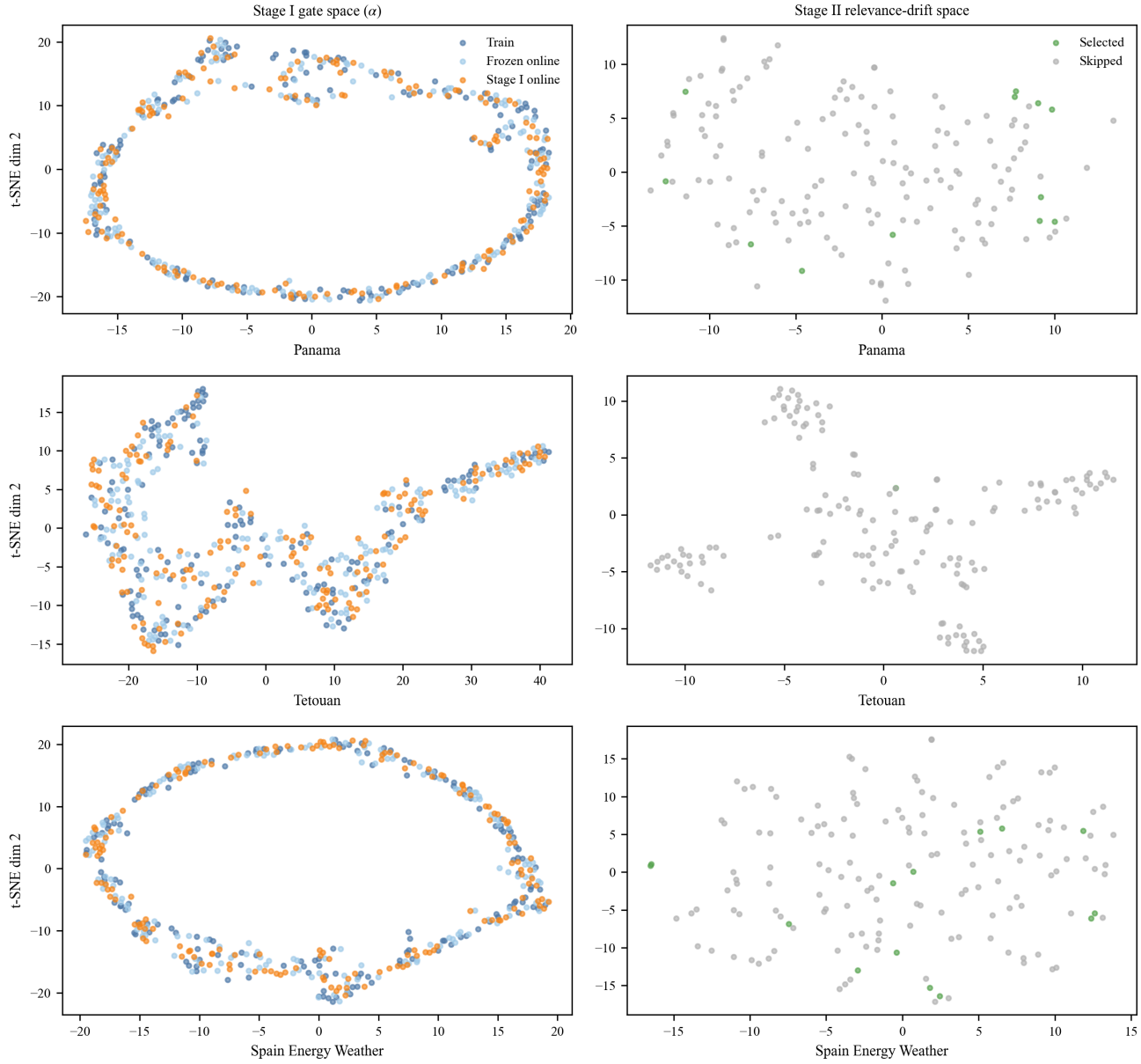


Fig. 9. t-SNE visualization of the Stage I gate space and the Stage II relevance-drift space. Stage I mainly induces a mild gate re-alignment of the online samples, while Stage II selects only a sparse subset of points with concentrated relevance-drift signatures.

still improves over the frozen family baseline after realistic load–weather alignment and cleaning, and it also reveals that the remaining gap to the strongest external controller is now concentrated in the online control policy rather than in the backbone accuracy itself. Overall, the experiments show that the proposed framework can simultaneously provide strong static forecasting accuracy and measurable online adaptation gains on multivariate regional load forecasting benchmarks.

18) Backbone Generality Discussion: A key design goal of the proposed method is to decouple the online adaptation mechanism from the specific forecasting backbone. The current results support this design choice. Once the temporal encoder is replaced, the remaining components—the invariant branch, the variant branch, the factor-VAE module, the expert mechanism layer, and the three online stages—can be reused without changing the overall training or deployment protocol. From this perspective, the role of the backbone is to provide a strong temporal representation, whereas the proposed framework contributes the mechanism decomposition and the online adaptation logic on top of that representation. The ranking in Table III therefore has a clear methodological interpretation: the framework is portable across multiple transformer-based predictors, while the final absolute performance still depends on the representational strength of the chosen backbone.

19) Online Adaptation Discussion: The all-backbone comparison in Table IV also clarifies the functional roles of the three online stages. Stage I adjusts the factor-weight predictor m_η using delayed labels, and is therefore most useful when the expert mechanisms remain valid but their mixture weights require recalibration. Stage II performs sparse relevance-guided updates only on unstable experts, so its effect depends more strongly on the quality of the backbone representation and

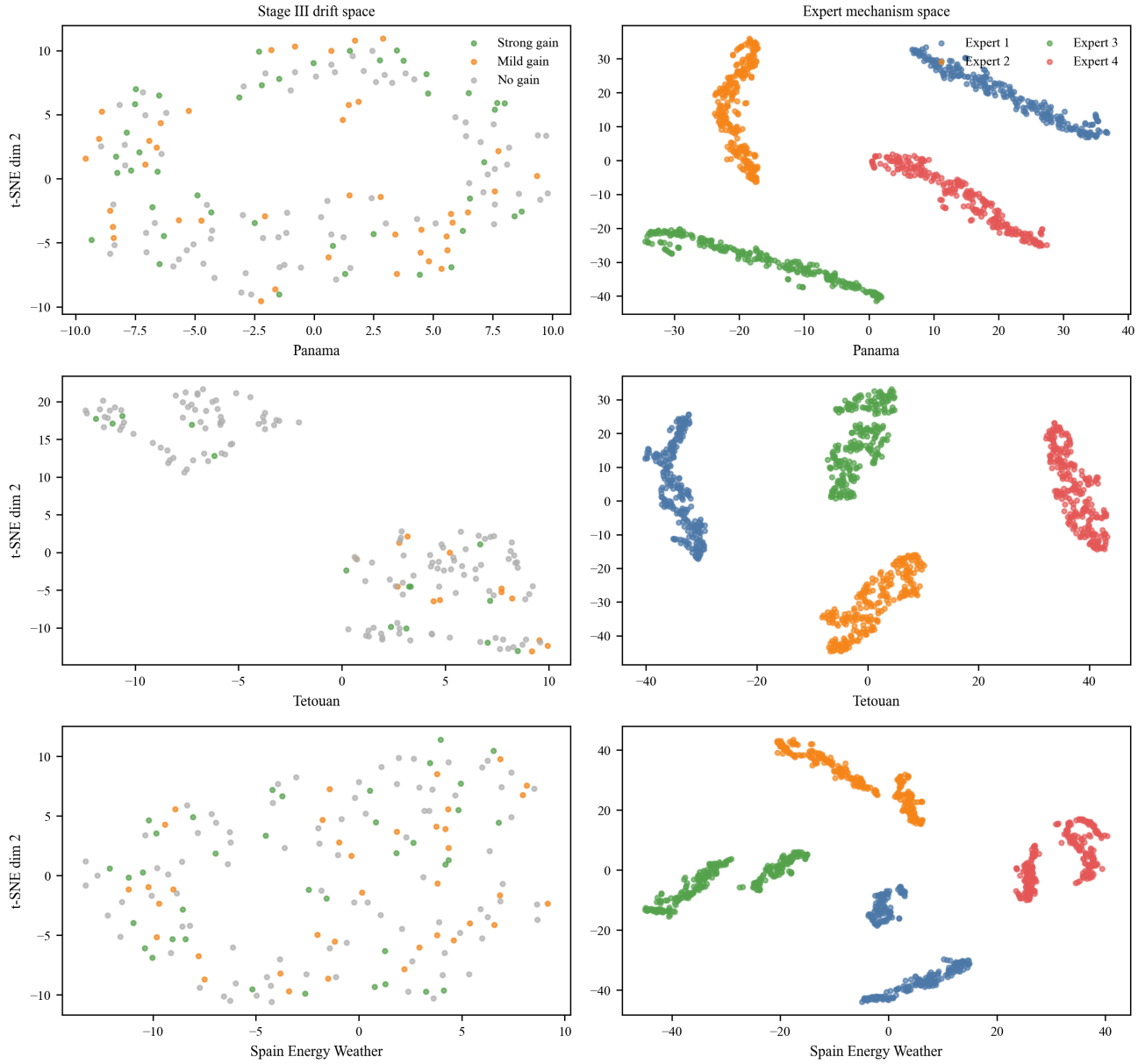


Fig. 10. t-SNE visualization of the Stage III correction space and the expert mechanism space. The left column colors online points by the realized gain of Stage III over the frozen prediction, while the right column shows that the expert embeddings form clearly separated clusters.

on the reliability of the relevance signal. Stage III uses the discrepancy $\phi_V - \phi_I$ to construct a transient correction at inference time, and is empirically the most stable source of online gain in the current benchmark suite. The combined model benefits from these three stages when their respective corrections are complementary rather than redundant, which explains why the strongest joint gains are observed on selected backbone–dataset combinations rather than uniformly across all settings.

20) Practical Implications: From a practical load-forecasting perspective, the current experiments suggest the following deployment strategy. When a strong backbone such as iTransformer or FEDformer is available, the proposed framework can be used directly to obtain both competitive static forecasting accuracy and additional online gains under delayed feedback. When computational or operational constraints are tighter, Stage III alone already provides a robust and lightweight improvement over the frozen baseline. When stronger online flexibility is allowed, enabling all three stages offers the best opportunity to exploit short-term drift, especially on backbones whose mechanism layer and relevance signal are sufficiently stable. This staged interpretation is important for practice because it allows the proposed method to be deployed under different latency, stability, and update-budget constraints without changing the overall framework.

TABLE VIII
QUANTITATIVE SHIFT-SCORE ANALYSIS ON THE THREE MAIN LOAD DATASETS. NUMBERS ARE ONLINE MSE ON SAMPLED SHIFTED TEST BLOCKS WITH EXACT SHIFT SCORES. LOWER IS BETTER.

Dataset	Shift Score	Frozen	Stage III	Full
Panama	0.0884	0.1883	0.1853	**0.1853**
Panama	0.1126	0.1873	**0.1833**	0.1834
Panama	0.1324	0.1983	**0.1951**	0.1953
Tetouan	0.3976	0.0953	0.0951	**0.0935**
Tetouan	0.4186	0.1037	0.1031	**0.1031**
Tetouan	0.4878	0.0849	**0.0836**	0.0838
Spain Energy–Weather	0.4547	0.2609	**0.2539**	0.2558
Spain Energy–Weather	0.4905	0.2571	0.2542	**0.2526**
Spain Energy–Weather	0.5396	0.2411	**0.2362**	0.2367

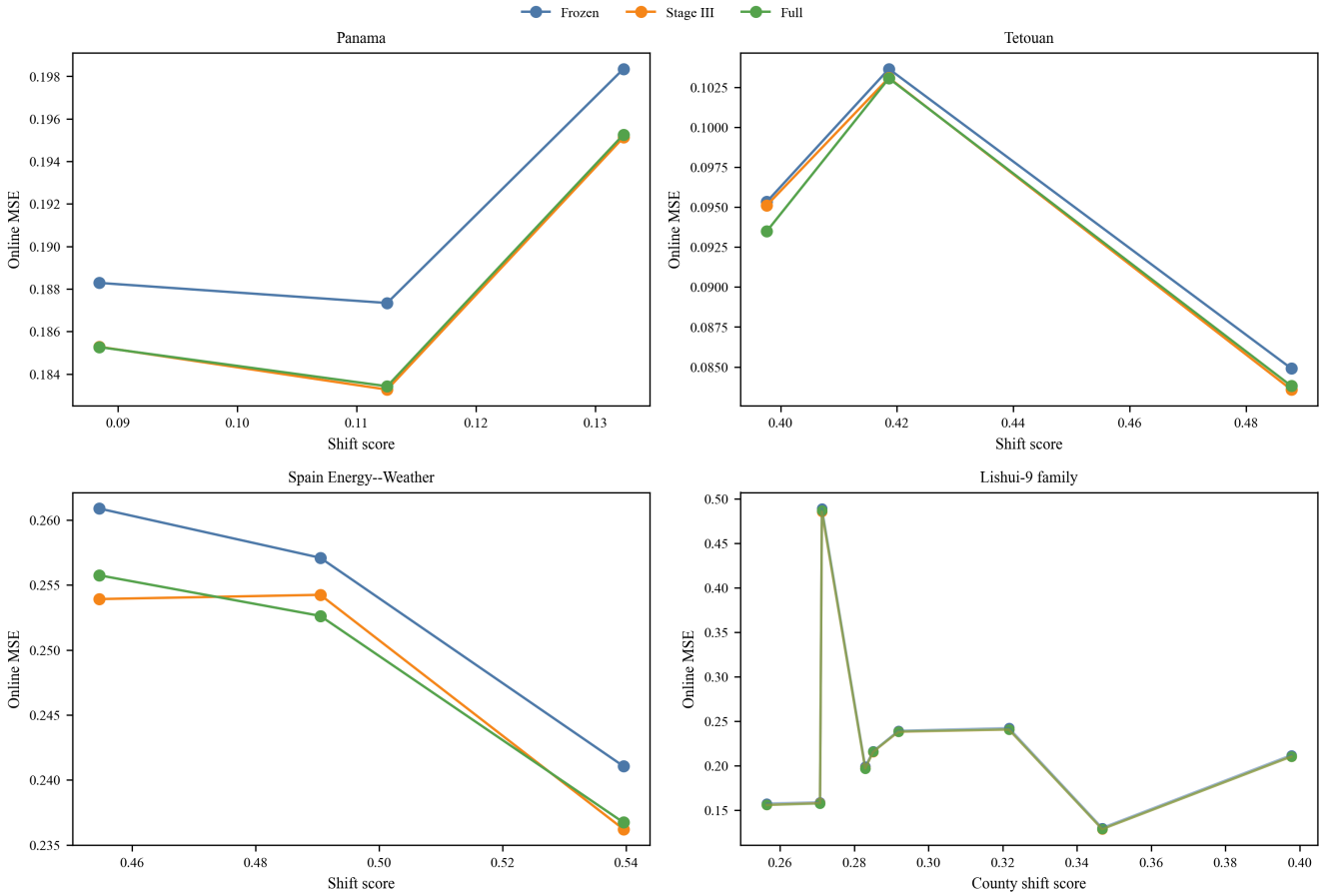


Fig. 11. Online MSE as a function of the exact shift score on the four datasets. The first three panels correspond to *Panama*, *Tetouan*, and *Spain Energy–Weather*; the fourth panel plots the nine Lishui counties individually.

TABLE IX
ADDITIONAL EVALUATION ON THE *Lishui-9* DATASET FAMILY. THE LEFT BLOCK REPORTS THE MEAN ONLINE MSE OF THE FULL EIGHT-VARIANT STAGE BENCHMARK OVER NINE COUNTIES. THE RIGHT BLOCK REPORTS THE MEAN ONLINE MSE OF THE TUNED SHIFTED-STREAM COMPARISON. LOWER IS BETTER.

Family benchmark									Tuned shifted-stream comparison				
Variant	Baseline	Stage I	Stage II	Stage III	Stage I+II	Stage I+III	Stage II+III	Full	Method	Frozen	SOLID++	Ours-Stage III	Ours-Full
Mean online MSE	0.2271	0.2269	0.2271	**0.2254**	0.2269	0.2256	**0.2254**	0.2256	Mean online MSE	0.2265	**0.2241**	0.2254	0.2255

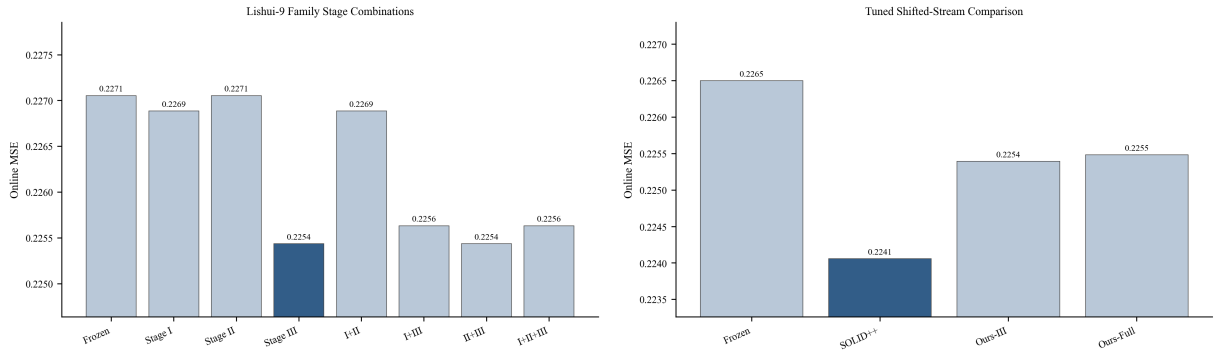


Fig. 12. Summary of the newly curated *Lishui-9* dataset family. Left: family-level mean online MSE of the stage combinations. Right: tuned shifted-stream comparison against **SOLID++**.

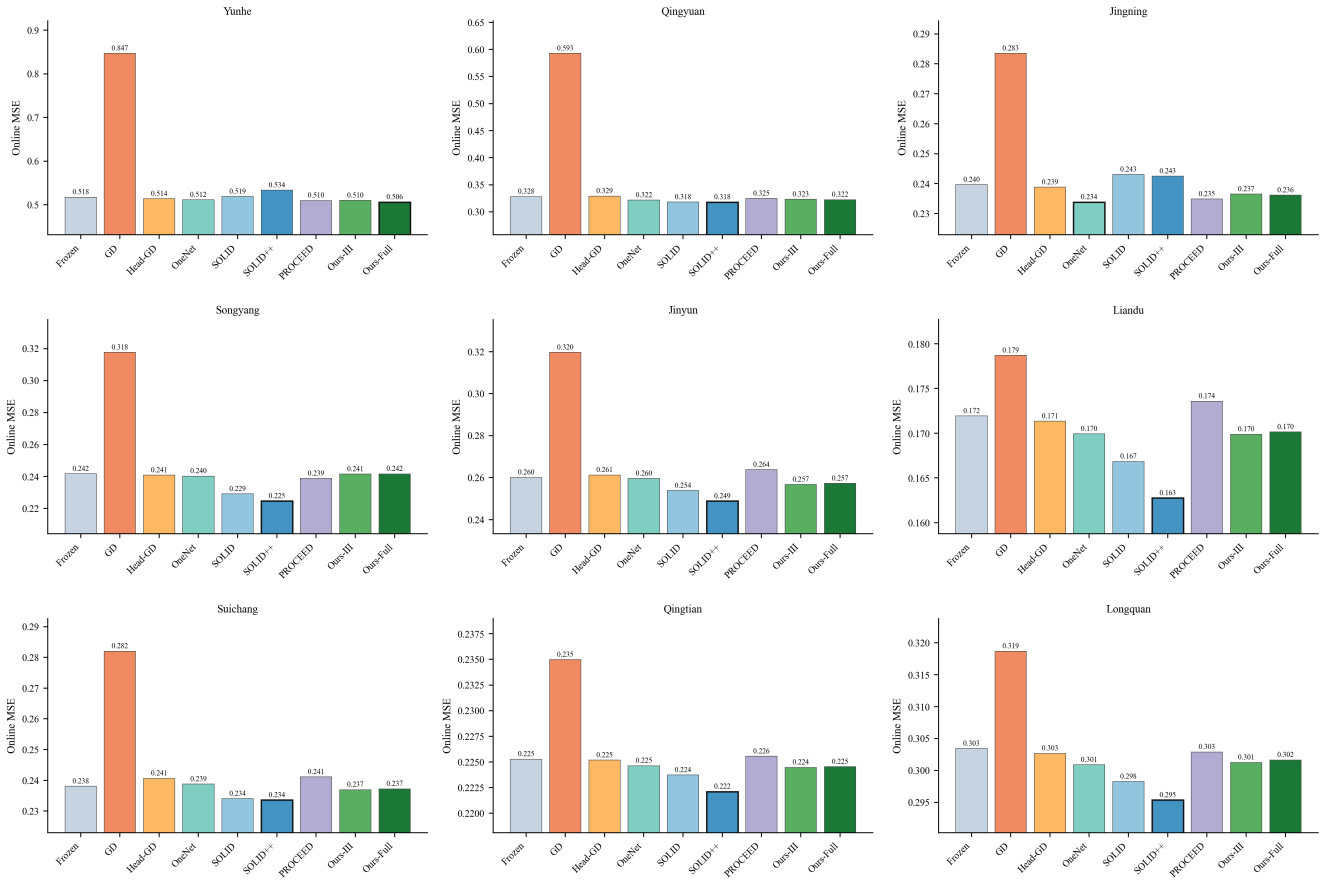


Fig. 13. County-level online comparison on the *Lishui-9* family under the shifted-stream protocol. Each subfigure shows one county and compares all online baselines and the proposed variants using online MSE.

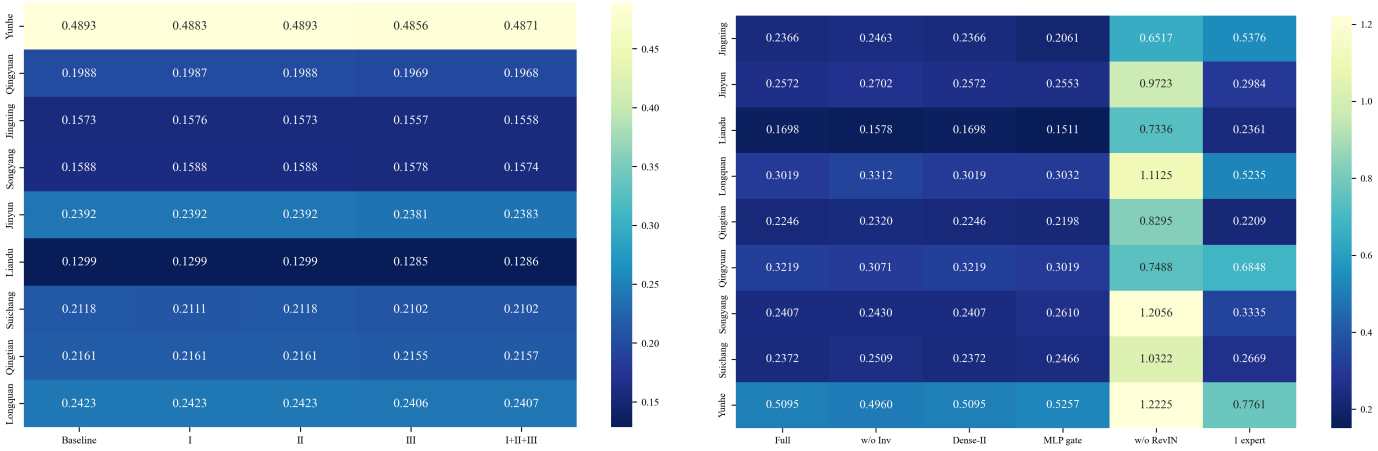


Fig. 14. County-level ablation on the *Lishui-9* family. Left: stage-combination benchmark over the full eight variants. Right: common structural component ablations.

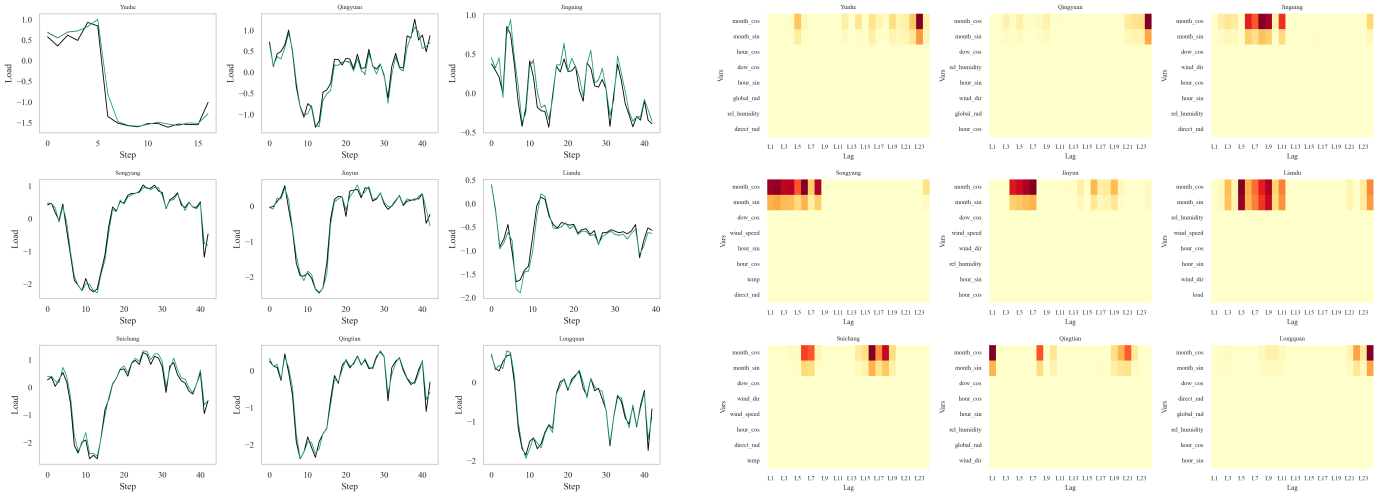


Fig. 15. County-level qualitative analysis on the *Lishui-9* family. Left: shifted-segment case studies for the nine counties using the same informative-window selection strategy as Fig. 6. Right: representative input-layer relevance maps for the same nine counties.

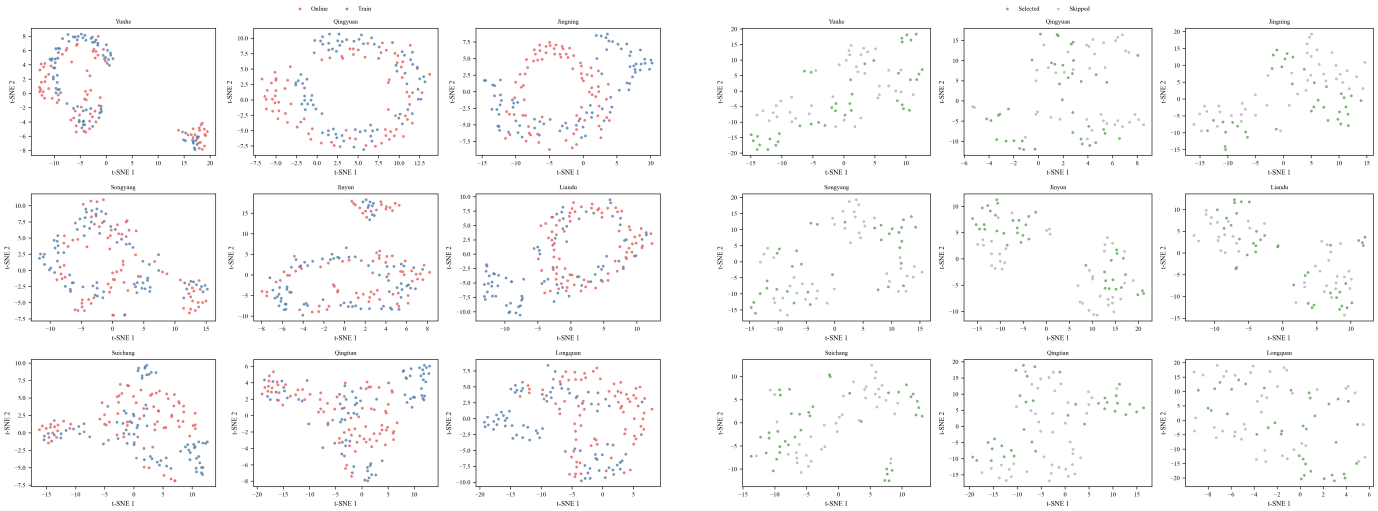


Fig. 16. County-level t-SNE visualization on the *Lishui-9* family. Left: train–online separation in the drift representation. Right: Stage II controller-space embedding for the nine counties.

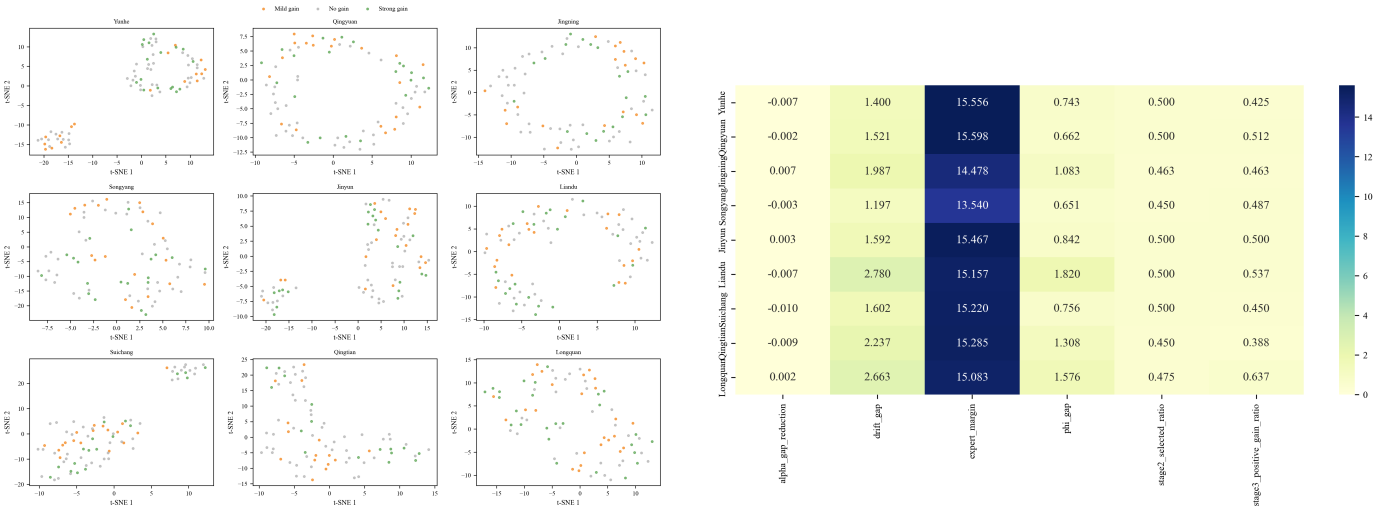


Fig. 17. Additional county-level t-SNE views on the *Lishui-9* family. Left: Stage III gain space across the nine counties. Right: county-level summary metrics extracted from the t-SNE analysis.

IV. Conclusion

This paper proposes a plug-and-play online load forecasting framework that combines invariant learning, factor-based mechanism modeling, and a three-stage online controller. The experiments show that the framework is compatible with multiple transformer-based backbones, consistently improves over frozen deployment under shifted online streams, and remains competitive against representative online forecasting baselines. The empirical evidence further indicates that Stage III provides the most reliable single-step gain, while the joint controller yields the strongest average performance when the online update policy is sufficiently constrained.

Acknowledgments

This section will be completed in the final manuscript.